EXAM PACK

# EEE329

Basic Communication Engineering



# Exam Pack

**Session 2021–22** *Solutions and formula sheet of the Third Year Second Semester Final Examination 2024*



## IN THIS PACK


1 PAST PAPER SOLUTIONS
Question Paper Solutions 2021–22 — pages 3--57 — 55 pages

2 FORMULA SHEET
Formula Sheet — pages 58--68 — 11 pages


Tap a document to open it. Each one keeps its own cover, contents and links.


Prepared by Nur Mohammad


October 2026

IN THIS PACK

# Exam Pack

Tap a heading to open that document, or a line to go straight to that page. Each document keeps its own cover, contents and links; the bookmarks panel lists them all.

## 1 PAST PAPER SOLUTIONS
## Question Paper Solutions 2021–22

pages 3--57


Question Paper . . . 5
1(a) Entropy of a Memoryless Source . . . 10
1(b) Modes of Propagation of EM Waves . . . 11
1(c) Attenuation, Channel Capacity and Required Signal Level . . . 13
1(d) Conditions for Distortionless Transmission . . . 15
2(a) AM Antenna Current and Percentage Modulation . . . 18
2(b) Effect of Phase and Frequency Error in Coherent Detection . . . 19
2(c) Costas Receiver for DSB-SC . . . 21
2(d) Condition on the VSB Shaping Filter . . . 22
3(a) Generation of NBFM . . . 28
3(b) Linear Diode (Envelope) Detector . . . 30
3(c) FM Carrier Swing, Frequency Limits and Modulation Index . . . 31
3(d) Sketch of the AM Signal for the Sawtooth Message . . . 32
4(a) Pulse Code Modulation . . . 36
4(b) Reconstruction of a Sampled Signal by an LPF 37
4(c) Short Notes: Companding, Information Rate, Entropy . . . 40
4(d) Digital-to-Analog Modulation: ASK, FSK and PSK . . . 41
4(e) PCM Signal-to-Quantization-Noise Ratio and Bandwidth . . . 43
5(a) Sigma-Delta Modulation and its Advantages over DM . . . 45
5(b) Coherent Detection of QAM . . . 46
5(c) DPSK Encoder and Receiver Table for ‘1010010110’ . . . 48
5(d) Coherent and Non-coherent Detection of BASK and BFSK . . . 49
6(a) Multilevel Multiplexing and Multiple-Slot Allocation . . . 52
6(b) Minimum Cable Bandwidth for the ECG TDM System . . . 53
6(c) FDMA versus CDMA . . . 54
6(d) Design Procedure of a Communication System 55
6(e) Multiplexing versus Multiple Access . . . 56


## 2 FORMULA SHEET
## Formula Sheet

pages 58--68


1 Information and channel capacity . . . 61
2 Channels and propagation . . . 62
3 Amplitude modulation . . . 62
4 Frequency modulation . . . 63
5 Sampling and PCM . . . 63
6 Digital modulation . . . 64
7 Multiplexing . . . 65
8 Key formulas . . . 66
9 Symbol index . . . 67

PAST PAPER SOLUTIONS

# EEE329

Basic Communication Engineering

## Question Paper Solutions

**Session 2021–22**

*EEE329 Third Year Second Semester Final Examination 2024*

QUESTION MAP

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 |
|---|---|---|---|---|---|
| 25 marks | 25 marks | 25 marks | 25 marks | 25 marks | 25 marks |

EACH ANSWER




Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

# The paper at a glance

| 3 h | 100 | 4 of 6 | 45 min |
|---|---|---|---|
| Total time | Total marks | Questions to answer | Per question |

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

| Q | Sub-part | Marks | Time | Pick |
|---|---|---|---|---|
| Part-A · Answer any two of the following questions | | | | |
| Q1 | (a) Entropy of a Memoryless Source | 6 | 11 min | ☐ |
| | (b) Modes of Propagation of EM Waves | 7 | 13 min | |
| | (c) Attenuation, Channel Capacity and Required Signal Level | 6 | 11 min | |
| | (d) Conditions for Distortionless Transmission | 6 | 11 min | |
| Q2 | (a) AM Antenna Current and Percentage Modulation | 5 | 9 min | ☐ |
| | (b) Effect of Phase and Frequency Error in Coherent Detection | 6 | 11 min | |
| | (c) Costas Receiver for DSB-SC | 8 | 14 min | |
| | (d) Condition on the VSB Shaping Filter | 6 | 11 min | |
| Q3 | (a) Generation of NBFM | 5 | 9 min | ☐ |
| | (b) Linear Diode (Envelope) Detector | 5 | 9 min | |
| | (c) FM Carrier Swing, Frequency Limits and Modulation Index | 5 | 9 min | |
| | (d) Sketch of the AM Signal for the Sawtooth Message | 10 | 18 min | |
| Part-B · Answer any two of the following questions | | | | |
| Q4 | (a) Pulse Code Modulation | 5 | 9 min | ☐ |
| | (b) Reconstruction of a Sampled Signal by an LPF | 5 | 9 min | |
| | (c) Short Notes: Companding, Information Rate, Entropy | 5 | 9 min | |
| | (d) Digital-to-Analog Modulation: ASK, FSK and PSK | 5 | 9 min | |
| | (e) PCM Signal-to-Quantization-Noise Ratio and Bandwidth | 5 | 9 min | |
| Q5 | (a) Sigma-Delta Modulation and its Advantages over DM | 6 | 11 min | ☐ |
| | (b) Coherent Detection of QAM | 7 | 13 min | |
| | (c) DPSK Encoder and Receiver Table for ‘1010010110’ | 6 | 11 min | |
| | (d) Coherent and Non-coherent Detection of BASK and BFSK | 6 | 11 min | |
| Q6 | (a) Multilevel Multiplexing and Multiple-Slot Allocation | 6 | 11 min | ☐ |
| | (b) Minimum Cable Bandwidth for the ECG TDM System | 6 | 11 min | |
| | (c) FDMA versus CDMA | 3 | 5 min | |
| | (d) Design Procedure of a Communication System | 7 | 13 min | |
| | (e) Multiplexing versus Multiple Access | 3 | 5 min | |

Formulas and conventions follow Lathi and Ding, Modern Digital and Analog Communication Systems (4th ed.); 1(b), 1(c), 4(d), 6(a), 6(c) and 6(e) follow Forouzan, Data Communications and Networking.

# Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

**B.Sc. (Engg.) Third Year, Second Semester Final Examination-2024**

**Course Code:** EEE 329 **Session:** 2021-2022

**Course Title:** Basic Communication Engineering

**Total Marks: 100** **Credits: 3.0** **Total Time: 3 Hours**

*[The figures in the right margin indicate full marks]*

## Part-A

*[Answer any two of the following questions]*

1. (a) Find the entropy $H(m)$ of a memoryless source. 6

   (b) Discuss the different modes of propagation of EM wave. 7

   (c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV.

   (i) Calculate the attenuation.
   (ii) Determine the maximum channel capacity.
   (iii) To achieve a data rate of 100 Mbit $s^{-1}$, find the required signal level. 6

   (d) Explain the conditions required for distortionless transmission. 6

2. (a) The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8 5

   (b) Explain how phase and frequency error affect the coherent detection of AM. 6

   (c) With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver. 8

   (d) Find the condition of sideband shaping filter for VSB modulation. 6

3. (a) Explain NBFM generation process with necessary block diagrams and equations. 5

   (b) Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling. 5

   (c) A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz.

   (i) Find the carrier swing of the FM signal.
   (ii) What are the highest and lowest frequencies attained by the frequency modulated signal?
   (iii) Calculate the modulation index for the wave. 5

   (d) For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$. 10



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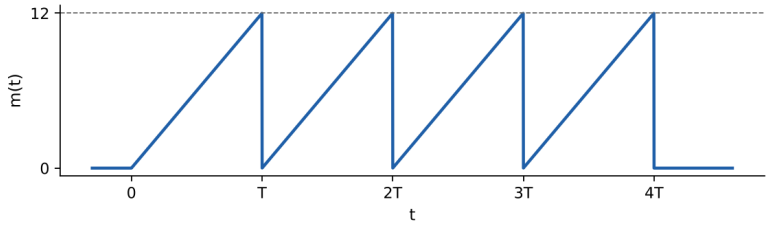



Fig. 1 of the paper (redrawn): the message m(t)

## Part-B

*[Answer any two of the following questions]*

4. (a) What is PCM? Explain PCM system. 5
   (b) "A proper valued LPF filter can be used to reconstruct the sampled signal"-justify the statement with the proper mathematical explanations. 5
   (c) Write short note on: Companding, Information rate, and Entropy of a source. 5
   (d) Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them. 5
   (e) An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find-
   (i) Peak signal power to peak quantization noise power.
   (ii) System bandwidth. 5

5. (a) Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM. 6
   (b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. 7
   (c) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'. 6
   (d) Briefly describe coherent and non-coherent detection of BASK and BFSK. 6

6. (a) Write short notes on: Multilevel multiplexing and Multiple slot allocation. 6
   (b) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. 6
   (c) Compare between FDMA and CDMA. 3
   (d) Discuss the design procedures of a communication system. 7
   (e) Differentiate between multiplexing and multiple access technologies. 3


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# Contents

**1(a) Entropy of a Memoryless Source** **8**
1 Theory . . . 8
2 Answer . . . 8
3 Keyword Explanation . . . 9
4 Notes and Remarks . . . 9

**1(b) Modes of Propagation of EM Waves** **9**
1 Theory . . . 9
2 Answer . . . 10
3 Notes and Remarks . . . 11

**1(c) Attenuation, Channel Capacity and Required Signal Level** **11**
1 Theory . . . 11
2 Answer . . . 12
3 Notes and Remarks . . . 13

**1(d) Conditions for Distortionless Transmission** **13**
1 Theory . . . 13
2 Answer . . . 13
3 Final Concise Answer . . . 15

**2(a) AM Antenna Current and Percentage Modulation** **16**
1 Theory . . . 16
2 Answer . . . 16
3 Notes and Remarks . . . 17

**2(b) Effect of Phase and Frequency Error in Coherent Detection** **17**
1 Theory . . . 17
2 Answer . . . 17
3 Notes and Remarks . . . 19

**2(c) Costas Receiver for DSB-SC** **19**
1 Theory . . . 19
2 Answer . . . 19
3 Keyword Explanation . . . 20

**2(d) Condition on the VSB Shaping Filter** **20**
1 Theory . . . 20
2 Answer . . . 21
3 Final Concise Answer . . . 25

**3(a) Generation of NBFM** **26**
1 Theory . . . 26
2 Answer . . . 26

**3(b) Linear Diode (Envelope) Detector** **28**
1 Theory . . . 28
2 Answer . . . 28
3 Notes and Remarks . . . 29

**3(c) FM Carrier Swing, Frequency Limits and Modulation Index** **29**


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1 Theory . . . 29
2 Answer . . . 29
3 Notes and Remarks . . . 30

**3(d) Sketch of the AM Signal for the Sawtooth Message** **30**
1 Theory . . . 30
2 Answer . . . 31
3 Notes and Remarks . . . 32
4 Final Concise Answer . . . 32

**4(a) Pulse Code Modulation** **34**
1 Theory . . . 34
2 Answer . . . 34

**4(b) Reconstruction of a Sampled Signal by an LPF** **35**
1 Theory . . . 35
2 Answer . . . 35

**4(c) Short Notes: Companding, Information Rate, Entropy** **38**
1 Theory . . . 38
2 Answer . . . 38

**4(d) Digital-to-Analog Modulation: ASK, FSK and PSK** **39**
1 Theory . . . 39
2 Answer . . . 39

**4(e) PCM Signal-to-Quantization-Noise Ratio and Bandwidth** **41**
1 Theory . . . 41
2 Answer . . . 41
3 Notes and Remarks . . . 42
4 Final Concise Answer . . . 42

**5(a) Sigma-Delta Modulation and its Advantages over DM** **43**
1 Theory . . . 43
2 Answer . . . 43

**5(b) Coherent Detection of QAM** **44**
1 Theory . . . 44
2 Answer . . . 45

**5(c) DPSK Encoder and Receiver Table for '1010010110'** **46**
1 Theory . . . 46
2 Answer . . . 46
3 Notes and Remarks . . . 47

**5(d) Coherent and Non-coherent Detection of BASK and BFSK** **47**
1 Theory . . . 47
2 Answer . . . 48
3 Final Concise Answer . . . 49

**6(a) Multilevel Multiplexing and Multiple-Slot Allocation** **50**
1 Theory . . . 50
2 Answer . . . 50


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**6(b) Minimum Cable Bandwidth for the ECG TDM System** **51**
1 Theory . . . 51
2 Answer . . . 51
3 Notes and Remarks . . . 52

**6(c) FDMA versus CDMA** **52**
1 Theory . . . 52
2 Answer . . . 53

**6(d) Design Procedure of a Communication System** **53**
1 Theory . . . 53
2 Answer . . . 53

**6(e) Multiplexing versus Multiple Access** **54**
1 Theory . . . 54
2 Answer . . . 55
3 Final Concise Answer . . . 55


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1. (a) Find the entropy $H(m)$ of a memoryless source. Marks: 6
   (b) Discuss the different modes of propagation of EM wave. Marks: 7
   (c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV.
      (i) Calculate the attenuation.
      (ii) Determine the maximum channel capacity.
      (iii) To achieve a data rate of 100 Mbit $s^{-1}$, find the required signal level. Marks: 6
   (d) Explain the conditions required for distortionless transmission. Marks: 6

**1.(a) Find the entropy $H(m)$ of a memoryless source. Marks: 6**

# 1. Theory

## 1. What is a source? (Core Idea)

A **source** is anything that produces messages: a keyboard, a microphone after sampling, a sensor. We model it as a machine that sends one **symbol** at a time, picked from a fixed list (the **alphabet**) $m_1, m_2, \ldots, m_n$. Symbol $m_i$ comes out with probability $P_i$, and all the probabilities add up to 1.

A source is **memoryless** when each symbol is chosen independently: what came before has no effect on what comes next. A fair coin is memoryless. English text is not (after "q" you almost always get "u").

## 2. How much information is in one symbol?

Think about news. "The sun rose today" tells you almost nothing, because you were sure of it. "It snowed in Dhaka" tells you a lot, because it was very unlikely. So **information grows when probability falls**. The measure that does this, and that adds up correctly for independent events, is:

$$I(m_i) = \log_2 \frac{1}{P_i} \text{ bits}$$

- A sure event ($P = 1$) carries $\log_2 1 = 0$ bits.
- One fair coin toss ($P = 1/2$) carries $\log_2 2 = 1$ bit. This is where the unit "bit" comes from.
- For two independent symbols, $P = P_1 P_2$, and the log turns the product into a sum: the information adds, as it should.

## 3. What is entropy?

One symbol may carry a lot of information and another very little. The **entropy** $H(m)$ is the *average* information per symbol, the number we need to design a system for the long run.

# 2. Answer

**Definition:** The entropy $H(m)$ of a memoryless source is the average information carried by one symbol of the source.

**Derivation:** Let the source emit symbols $m_1, \ldots, m_n$ with probabilities $P_1, \ldots, P_n$ ($\sum P_i = 1$). The information in symbol $m_i$ is

$$I_i = \log_2 \frac{1}{P_i} \text{ bits}$$

Take a long message of $N$ symbols. Since the source is memoryless, symbol $m_i$ appears about $NP_i$ times, and each time it brings $I_i$ bits. So the total information in the message is

$$I_{total} = \sum_{i=1}^{n} (NP_i)\, I_i = N \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$$


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Dividing by the number of symbols $N$ gives the average information per symbol:

$$\boxed{H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i} = -\sum_{i=1}^{n} P_i \log_2 P_i \quad \text{bits/symbol}}$$

**Key Points:**

- **Lowest value:** $H = 0$ when one symbol is certain ($P_k = 1$, all others 0). A source that always says the same thing tells us nothing.
- **Highest value:** $H = \log_2 n$ when all $n$ symbols are equally likely ($P_i = 1/n$). Then $H = \sum \frac{1}{n} \log_2 n = \log_2 n$. So $0 \le H(m) \le \log_2 n$.
- **Binary source** (two symbols, probabilities $p$ and $1-p$):

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

It is 0 at $p = 0$ or $p = 1$ and reaches its maximum of 1 bit at $p = 0.5$. For example $H(0.1) = H(0.9) = 0.469$ bit.

**Worked example**

A memoryless source has four symbols with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$. Then $H = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = 0.5 + 0.5 + 0.375 + 0.375 = 1.75$ bits/symbol, which is less than the maximum $\log_2 4 = 2$ bits because the symbols are not equally likely.

**Interpretation:** $H(m)$ is the smallest average number of binary digits per symbol that any code can use and still describe the source without loss (Shannon's source coding theorem).

# 3. Keyword Explanation

## 1. Bit (unit of information)

The information in one choice between two equally likely outcomes. With $\log_2$ the unit is the bit; with the natural log it is the nat.

## 2. Memoryless (zero-memory) source

A source whose symbols are statistically independent: the probability of the next symbol does not depend on earlier symbols.

# 4. Notes and Remarks

**Why is the maximum $\log_2 n$?** Using $\ln x \le x - 1$:

$$H - \log_2 n = \sum_i P_i \log_2 \frac{1}{nP_i} \le \frac{1}{\ln 2} \sum_i P_i \left( \frac{1}{nP_i} - 1 \right) = \frac{1}{\ln 2} (1 - 1) = 0$$

so $H \le \log_2 n$, with equality only when every $\frac{1}{nP_i} = 1$, that is $P_i = 1/n$.

**1.(b) Discuss the different modes of propagation of EM wave. Marks: 7**

# 1. Theory

## 1. What is an EM wave? (Core Idea)

A radio signal leaves the antenna as an **electromagnetic (EM) wave**: electric and magnetic fields that travel at the speed of light, about $3 \times 10^8$ m/s. Between the transmitter and the receiver there are only three "roads" the wave can take: along the ground, up to the sky and back, or straight through the air. Which road works best depends mainly on the **frequency**.

### 2. Why does frequency decide the road?

- Low-frequency waves are long (at 300 kHz, $\lambda = c/f = 1$ km). Long waves bend around the curved Earth and over hills (**diffraction**), so they can follow the ground.
- Medium-high frequencies (about 2–30 MHz) are bent back to Earth by the **ionosphere**, a layer of charged gas 60–400 km up.
- Very high frequencies pass straight through the ionosphere into space, so the only way is a straight line from antenna to antenna.

## 2. Answer

EM waves travel from transmitter to receiver in three main modes: **ground (surface) wave**, **sky wave** and **line-of-sight (space) wave** propagation.

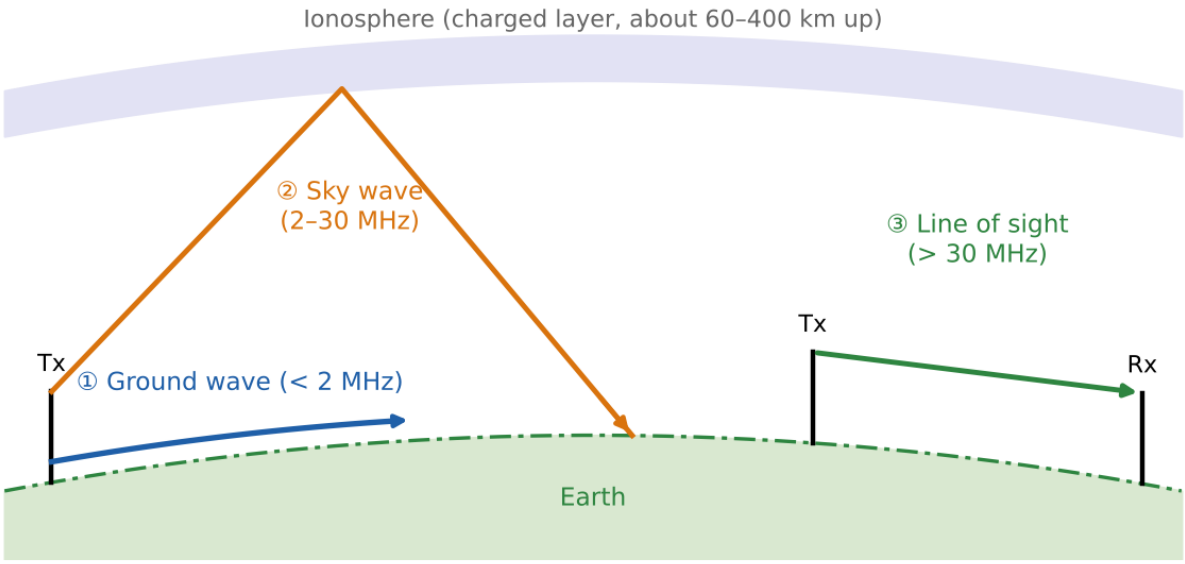



**Figure 1:** Three modes of propagation of EM waves

**Reading the figure.** The blue wave on the left (labelled 1) hugs the curved Earth (ground wave). The orange wave (labelled 2) goes up, is bent back by the ionosphere and lands far away (sky wave). The green wave on the right (labelled 3) goes straight from the Tx tower to the Rx tower (line of sight).

### 1. Ground (surface) wave propagation

- **Frequency:** below about 2 MHz (VLF, LF, MF).
- **How:** the wave travels close to the Earth's surface and follows its curvature. The lower part of the wavefront moves in the ground and is slowed down, so the wave tilts and bends with the Earth.
- **Range:** depends on transmitter power; can be hundreds of km. The wave loses energy into the ground, and the loss grows with frequency.
- **Uses:** AM (MW) broadcasting, maritime and navigation signals.
- **Needs:** vertical polarization and large antennas (antenna size is comparable to $\lambda$).

### 2. Sky wave (ionospheric) propagation

- **Frequency:** about 2–30 MHz (HF, short wave).
- **How:** the wave goes upward and is refracted (bent back) by the ionized layers of the ionosphere, returning to Earth far from the transmitter. Several hops between ground and ionosphere give very long range.
- **Range:** thousands of km, even around the world, with low power.
- **Uses:** short-wave international broadcasting, amateur radio, long-distance ship and aircraft links.
- **Drawbacks:** the ionosphere changes with day, night and season, so signals fade; there is a **skip zone** where neither the ground wave nor the sky wave arrives.

### 3. Line-of-sight (space wave) propagation

- **Frequency:** above about 30 MHz (VHF, UHF, microwave).
- **How:** the wave travels in a straight line from the transmitting antenna to the receiving antenna (directly, or after one reflection from the ground). The ionosphere does not return it.
- **Range:** limited by the curvature of the Earth, so antennas are placed high. With the usual $\frac{4}{3}$-Earth-radius correction for atmospheric bending, the radio horizon is

$$d \approx 4.12 \left( \sqrt{h_t} + \sqrt{h_r} \right) \text{ km}$$

where:

- $h_t$, $h_r$ = heights of the transmitting and receiving antennas in metres.
- **Uses:** FM radio, TV, mobile phones, microwave links, satellite communication (where the satellite is in line of sight).

| Mode | Frequency | Path | Range | Example |
|---|---|---|---|---|
| Ground wave | $< 2$ MHz | follows Earth's surface | up to a few hundred km | AM broadcast |
| Sky wave | 2–30 MHz | reflected by ionosphere | thousands of km | short-wave radio |
| Line of sight | $> 30$ MHz | straight line | horizon (tens of km) or satellite | FM, TV, mobile |

**Table 1:** Comparison of the three propagation modes

## 3. Notes and Remarks

Forouzan's band table puts MF (300 kHz–3 MHz) under sky propagation and VHF under both sky and line-of-sight propagation. Both views are right: an AM (MF) broadcast travels by ground wave in the day and also by sky wave at night, when the lowest ionospheric layer disappears. The boundaries 2 MHz and 30 MHz used above are approximate.

Some books add **tropospheric scatter** (UHF signals scattered by the lower atmosphere, a few hundred km) and **satellite** links (a line-of-sight path via a satellite). They are special cases of the three modes above.

**1.(c) We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV. (i) Calculate the attenuation. (ii) Determine the maximum channel capacity. (iii) To achieve a data rate of 100 Mbit $s^{-1}$, find the required signal level. Marks: 6**

## 1. Theory

### 1. Attenuation (Core Idea)

**Attenuation** is the loss of signal strength along the channel. It is measured as the ratio of transmitted to received power, in decibels:

$$\text{Attenuation (dB)} = 10 \log_{10} \frac{P_t}{P_r} = 20 \log_{10} \frac{V_t}{V_r}$$

The second form uses voltage amplitudes. Power is proportional to $V^2$ (for the same resistance), and $10 \log V^2 = 20 \log V$.

### 2. Signal-to-noise ratio

The receiver sees both the signal and noise. Their power ratio is the **SNR**:

$$\text{SNR} = \frac{S}{N} = \left(\frac{V_s}{V_n}\right)^2$$

### 3. Shannon capacity: the speed limit of a channel

Shannon proved that a channel of bandwidth $B$ with signal-to-noise ratio $S/N$ can carry at most

$$C = B \log_2\left(1 + \frac{S}{N}\right) \text{ bit/s}$$

without errors. More bandwidth or a stronger signal raises the limit; no clever coding can beat it. Note that $S/N$ here is the plain ratio, not dB.

## 2. Answer

**Given:**

- Bandwidth $B = 5$ MHz $= 5 \times 10^6$ Hz
- Transmitted amplitude $V_t = 10$ V
- Received signal amplitude $V_r = 2$ V
- Noise amplitude $V_n = 5$ mV $= 0.005$ V
- Required data rate $R = 100$ Mbit/s $= 10^8$ bit/s

**Assumption:** signal and noise are measured across the same resistance, so power ratios are squared amplitude ratios.

### Step 1: Attenuation

$$\text{Attenuation} = 20 \log_{10} \frac{V_t}{V_r} = 20 \log_{10} \frac{10}{2} = 20 \log_{10} 5 = 13.98 \text{ dB}$$

$$\boxed{\text{Attenuation} \approx 13.98 \text{ dB}}$$

The voltage drops 5 times, so the power drops 25 times.

### Step 2: Maximum channel capacity

$$\frac{S}{N} = \left(\frac{V_r}{V_n}\right)^2 = \left(\frac{2}{0.005}\right)^2 = 400^2 = 160000 \quad (52.04 \text{ dB})$$

$$C = B \log_2\left(1 + \frac{S}{N}\right) = 5 \times 10^6 \times \log_2(160001) = 5 \times 10^6 \times 17.288 = 86.44 \times 10^6 \text{ bit/s}$$

$$\boxed{C \approx 86.44 \text{ Mbit/s}}$$

No coding scheme can send more than about 86.4 Mbit/s error-free through this channel.

### Step 3: Signal level for 100 Mbit/s

Set $C = R$ and solve Shannon's formula for $S/N$:

$$\log_2\left(1 + \frac{S}{N}\right) = \frac{R}{B} = \frac{10^8}{5 \times 10^6} = 20 \quad \Rightarrow \quad \frac{S}{N} = 2^{20} - 1 = 1048575 \quad (60.21 \text{ dB})$$

The noise stays at 5 mV, so the received signal amplitude must be

$$V_r' = V_n \sqrt{S/N} = 0.005 \times \sqrt{1048575} = 0.005 \times 1024 = 5.12 \text{ V}$$

$$\boxed{V_r' \approx 5.12 \text{ V at the receiver}}$$

With the same 13.98 dB attenuation (a factor of 5 in amplitude), the transmitter must send $5 \times 5.12 = 25.6$ V.

**Final Answer:**


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- (i) Attenuation = 13.98 dB.
- (ii) $C \approx 86.44$ Mbit/s.
- (iii) Required $S/N = 1048575$ (60.2 dB): received signal $\approx 5.12$ V (transmitted signal $\approx 25.6$ V).

# 3. Notes and Remarks

- The needed SNR rises by $60.21 - 52.04 = 8.17$ dB to go from 86.4 to 100 Mbit/s. Because capacity grows only with the *log* of SNR, each extra bit/s per Hz costs about 3 dB more signal power.
- Forouzan writes a loss as a negative decibel value, $10\log_{10}(P_r/P_t) = -13.98$ dB. The size is the same; "an attenuation of 13.98 dB" and "a gain of $-13.98$ dB" mean the same thing.
- If the question's "signal level" is read as the transmitted level, the answer is 25.6 V; as the received level, 5.12 V. Both are given above.

**1.(d) Explain the conditions required for distortionless transmission. Marks: 6**

# 1. Theory

## 1. What does "distortionless" mean? (Core Idea)

A channel is **distortionless** if the output has exactly the same *shape* as the input. Two changes are allowed because they do not change the shape: the output may be **bigger or smaller** (a constant gain $k$) and it may arrive **later** (a constant delay $t_d$). So

$$y(t) = k\,x(t - t_d)$$

## 2. Why look at frequency?

Any signal is a sum of sinusoids of different frequencies. The shape is kept only if *every* sinusoid gets the *same* gain and the *same* time delay. If high frequencies are reduced more than low ones, or arrive later, the parts no longer add up to the original shape.

# 2. Answer

## Step 1: What the output must look like

A distortionless channel may only scale the signal by a constant $k$ and delay it by a constant time $t_d$ (Theory 1), so its output must be

$$y(t) = k\,x(t - t_d)$$

## Step 2 (Basic): The time-shift property of the Fourier transform

Take the Fourier transform of a delayed signal $x(t - t_d)$, from the definition $X(f) = \int_{-\infty}^{\infty} x(t)\,e^{-j2\pi ft}\,dt$:

$$\mathcal{F}\{x(t - t_d)\} = \int_{-\infty}^{\infty} x(t - t_d)\,e^{-j2\pi ft}\,dt$$

Substitute $\tau = t - t_d$, so $t = \tau + t_d$ and $dt = d\tau$ (the limits stay $-\infty$ to $\infty$):

$$\int_{-\infty}^{\infty} x(\tau)\,e^{-j2\pi f(\tau + t_d)}\,d\tau = e^{-j2\pi ft_d}\int_{-\infty}^{\infty} x(\tau)\,e^{-j2\pi f\tau}\,d\tau = e^{-j2\pi ft_d}\,X(f)$$

The factor $e^{-j2\pi ft_d}$ does not depend on $\tau$, so it comes out of the integral, and what is left is $X(f)$. So a delay of $t_d$ multiplies the spectrum by $e^{-j2\pi ft_d}$:

$$x(t - t_d) \Longleftrightarrow X(f)\,e^{-j2\pi ft_d}$$

## Step 3: The transfer function of the channel

Take the Fourier transform of Step 1. The constant $k$ stays outside, and Step 2 handles the delay:

$$Y(f) = k\,X(f)\,e^{-j2\pi ft_d}$$


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The transfer function is the output spectrum divided by the input spectrum:

$$\boxed{H(f) = \frac{Y(f)}{X(f)} = k\, e^{-j2\pi f t_d}}$$

## Step 4: Read off the magnitude and the phase

A complex number $e^{-j\phi}$ has magnitude 1 and angle $-\phi$ (since $e^{-j\phi} = \cos\phi - j\sin\phi$ and $\cos^2\phi + \sin^2\phi = 1$). So $H(f) = k\, e^{-j2\pi f t_d}$ has magnitude $k$ and angle $-2\pi f t_d$. This gives two conditions, over the whole band of the signal:

1. **Constant magnitude (amplitude) response:**

$$|H(f)| = k \quad \text{(the same for all frequencies)}$$

Every frequency component is multiplied by the same gain.

2. **Linear phase response:**

$$\theta_h(f) = -2\pi f t_d$$

The phase is a straight line through the origin with slope $-2\pi t_d$. Then every component is delayed by the same time $t_d$. (A phase of $-2\pi f t_d \pm n\pi$ only flips the sign and is also acceptable.)

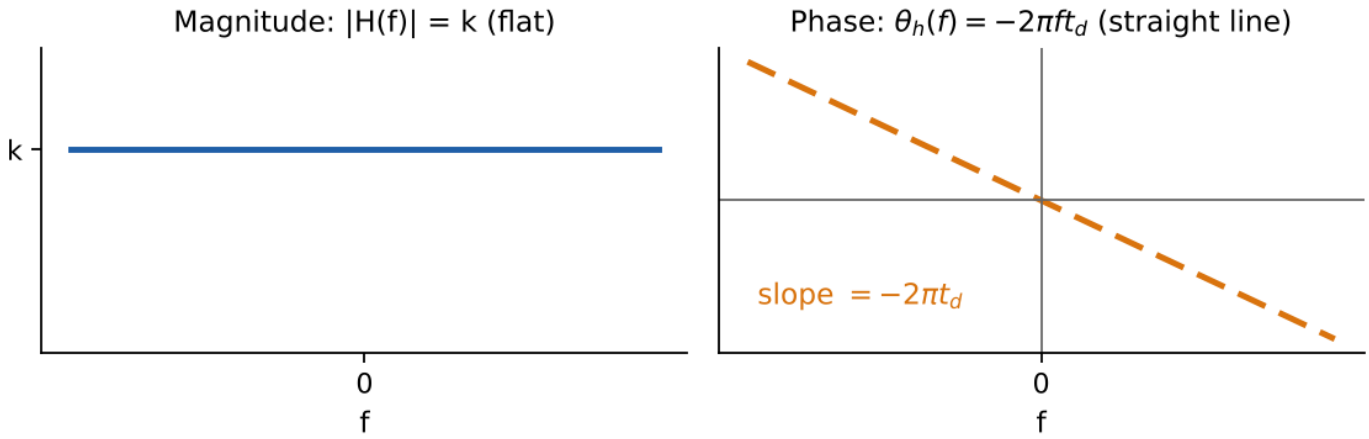



**Figure 2:** Magnitude and phase response of a distortionless channel

**Reading the figure.** The left plot is the magnitude, a flat line at height k. The right plot is the phase, a straight line through the origin whose slope is $-2\pi t_d$.

**Why linear phase gives equal delay:** a sinusoid of frequency $f$ that gets a phase $\theta$ can be rewritten as a delayed sinusoid:

$$\cos(2\pi f t + \theta) = \cos\left[2\pi f\left(t + \frac{\theta}{2\pi f}\right)\right] = \cos[2\pi f(t - t_f)], \qquad t_f = -\frac{\theta}{2\pi f}$$

With $\theta = \theta_h(f) = -2\pi f t_d$, the delay is $t_f = -(-2\pi f t_d)/(2\pi f) = t_d$, the same for every $f$. If the phase were not a straight line through the origin, $t_f$ would change with $f$, and the components would arrive at different times. Equivalently, the **group delay** $t_g = -\frac{1}{2\pi}\frac{d\theta_h}{df} = -\frac{1}{2\pi}(-2\pi t_d) = t_d$ is constant.

**Types of distortion when the conditions fail:**

- **Amplitude distortion:** $|H(f)|$ is not flat, so some frequencies are amplified more than others.
- **Phase (delay) distortion:** $\theta_h(f)$ is not linear, so different frequencies are delayed by different times. The ear tolerates it in speech, but it smears pulses in data and video.
- **Non-linear distortion:** the channel itself is non-linear (e.g. an overdriven amplifier), creating new frequencies; no $H(f)$ describes it.

**Practical Note:** a real channel only needs to meet the two conditions over the band occupied by the signal. Where it does not, an **equalizer** with the inverse response is added at the receiver.

# 3. Final Concise Answer

**(a)** For a memoryless source with symbols of probabilities $P_i$: $H(m) = \sum P_i \log_2(1/P_i)$ bits/symbol, with $0 \leq H \leq \log_2 n$ (maximum for equally likely symbols; 1 bit for a fair binary source).

**(b)** Three modes: ground wave ($< 2$ MHz, follows the Earth, AM broadcast), sky wave (2–30 MHz, bent back by the ionosphere, very long range), line of sight ($> 30$ MHz, straight path, range $\approx 4.12(\sqrt{h_t} + \sqrt{h_r})$ km; FM, TV, mobile, satellite).

**(c)** Attenuation $= 20\log(10/2) = 13.98$ dB; $S/N = 160000$, $C = 5\text{M} \times \log_2(160001) \approx 86.44$ Mbit/s; for 100 Mbit/s, $S/N = 2^{20} - 1$, so the received signal must be $\approx 5.12$ V (25.6 V at the transmitter).

**(d)** $y(t) = kx(t - t_d)$, so $H(f) = ke^{-j2\pi f t_d}$: constant magnitude $|H(f)| = k$ and linear phase $\theta_h = -2\pi f t_d$ (constant delay) over the signal band.


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2. (a) The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8 Marks: 5
   (b) Explain how phase and frequency error affect the coherent detection of AM. Marks: 6
   (c) With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver. Marks: 8
   (d) Find the condition of sideband shaping filter for VSB modulation. Marks: 6

**2.(a) The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8 Marks: 5**

# 1. Theory

## 1. What happens to power in AM? (Core Idea)

In AM the carrier's amplitude follows the message. For a single tone,

$$\varphi_{AM}(t) = A_c[1 + \mu \cos \omega_m t] \cos \omega_c t = A_c \cos \omega_c t + \mu A_c \cos \omega_m t \cos \omega_c t$$

Use $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ on the second term, with $A = \omega_c t$ and $B = \omega_m t$:

$$\varphi_{AM}(t) = \underbrace{A_c \cos \omega_c t}_{\text{carrier}} + \underbrace{\frac{\mu A_c}{2} \cos(\omega_c - \omega_m)t}_{\text{lower sideband}} + \underbrace{\frac{\mu A_c}{2} \cos(\omega_c + \omega_m)t}_{\text{upper sideband}}$$

So AM is three sinusoids: the carrier and two **sidebands** at $f_c \pm f_m$, each with amplitude $\mu A_c/2$.

A sinusoid of peak $A$ has power $A^2/2$ (in a 1 Ω resistance: its rms value is $A/\sqrt{2}$). Sinusoids of different frequencies do not interfere on average, so the total power is the sum of the three powers:

$$P_t = \frac{A_c^2}{2} + 2 \cdot \frac{(\mu A_c/2)^2}{2} = \frac{A_c^2}{2} + \frac{\mu^2 A_c^2}{4} = \frac{A_c^2}{2}\left(1 + \frac{\mu^2}{2}\right) = P_c\left(1 + \frac{\mu^2}{2}\right)$$

where $P_c = A_c^2/2$ is the carrier power. So modulation *adds* power; the carrier power itself does not change.

## 2. From power to current

The antenna is a resistance $R$, and power $= I^2 R$. Since $R$ is the same with and without modulation:

$$\frac{I_t^2}{I_c^2} = \frac{P_t}{P_c} = 1 + \frac{\mu^2}{2} \quad \Rightarrow \quad I_t = I_c\sqrt{1 + \frac{\mu^2}{2}}$$

# 2. Answer

**Given:**

- Carrier-only (unmodulated) current $I_c = 8$ A
- Current with modulation $I_t = 8.93$ A
- New modulation index $\mu' = 0.8$ (80%)

**Relationship:**

$$I_t = I_c\sqrt{1 + \frac{\mu^2}{2}}$$

where:


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- $I_t$ = rms antenna current with modulation, $I_c$ = rms current of the carrier alone, $\mu$ = modulation index.

### Step 1: Percentage modulation

$$\left(\frac{I_t}{I_c}\right)^2 = 1 + \frac{\mu^2}{2} \quad \Rightarrow \quad \mu^2 = 2\left[\left(\frac{8.93}{8}\right)^2 - 1\right] = 2\,(1.24601 - 1) = 0.49203$$

$$\mu = \sqrt{0.49203} = 0.7015$$

$$\boxed{\mu \approx 0.70 \;\Rightarrow\; \text{percentage modulation} \approx 70.15\%}$$

### Step 2: Antenna current at $\mu = 0.8$

$$I_t' = I_c\sqrt{1 + \frac{\mu'^2}{2}} = 8\sqrt{1 + \frac{0.8^2}{2}} = 8\sqrt{1.32} = 8 \times 1.1489 = 9.19 \text{ A}$$

$$\boxed{I_t' \approx 9.19 \text{ A}}$$

More modulation puts more power into the sidebands, so the antenna current rises.
**Final Answer:** percentage modulation $\approx 70\%$ ($\mu = 0.7015$); at 80% modulation the antenna current is $\approx 9.19$ A.

## 3. Notes and Remarks

- The question writes "percentage of modulation changes to 0.8"; this is read as $\mu = 0.8$, i.e. 80%.
- Even at 100% modulation ($\mu = 1$) the current rises only to $8\sqrt{1.5} = 9.80$ A: at most one third of the total power is in the sidebands. This is why AM is power-inefficient.

**2.(b) Explain how phase and frequency error affect the coherent detection of AM. Marks: 6**

## 1. Theory

### 1. What is coherent detection? (Core Idea)

**Coherent** (synchronous) detection recovers the message by multiplying the received signal by a **local carrier** made at the receiver, then low-pass filtering. It works perfectly only if the local carrier has *exactly* the same frequency and phase as the transmitter's carrier. The receiver must make that carrier itself, so small errors are unavoidable. The question is: what do they do to the output?

### 2. The perfect case

Received DSB-SC signal $m(t)\cos\omega_c t$, local carrier $2\cos\omega_c t$:

$$m(t)\cos\omega_c t \cdot 2\cos\omega_c t = m(t) + m(t)\cos 2\omega_c t$$

The LPF removes the $2\omega_c$ term and leaves $m(t)$.

## 2. Answer

### Step 1: The received signal and the imperfect local carrier

The received DSB-SC signal is $m(t)\cos\omega_c t$. Let the local carrier have a frequency error $\Delta\omega$ and a phase error $\theta$: $2\cos[(\omega_c + \Delta\omega)t + \theta]$.

### Step 2: The product

$$e(t) = m(t)\cos\omega_c t \cdot 2\cos[(\omega_c + \Delta\omega)t + \theta]$$

Use $2\cos A\cos B = \cos(A-B) + \cos(A+B)$ with $A = \omega_c t$ and $B = (\omega_c + \Delta\omega)t + \theta$:

- $A - B = \omega_c t - \omega_c t - \Delta\omega\, t - \theta = -(\Delta\omega\, t + \theta)$, and $\cos(-x) = \cos x$, so $\cos(A-B) = \cos(\Delta\omega\, t + \theta)$.

- $A + B = (2\omega_c + \Delta\omega)t + \theta$.

$$e(t) = m(t)\cos(\Delta\omega\, t + \theta) + m(t)\cos[(2\omega_c + \Delta\omega)t + \theta]$$

## Step 3: The low-pass filter

The second term is the message moved up to about $2\omega_c$, far above the message band, so the LPF removes it. The first term is near baseband ($\Delta\omega$ is small) and passes. The output is

$$\boxed{e_o(t) = m(t)\cos(\Delta\omega\, t + \theta)}$$

## Case 1: Phase error only ($\Delta\omega = 0$)

$$e_o(t) = m(t)\cos\theta$$

- The output is the right message, only scaled by $\cos\theta$. A constant small error just makes it a little weaker.
- At $\theta = \pm 90°$ the output is *zero* (quadrature null effect).
- If $\theta$ drifts randomly with time (e.g. channel delay changes), the gain $\cos\theta$ keeps changing, so the output fades in and out.

## Case 2: Frequency error only ($\theta = 0$)

$$e_o(t) = m(t)\cos(\Delta\omega\, t)$$

- The message is multiplied by a slow sinusoid of frequency $\Delta f$. Its loudness rises and falls periodically (a **beat**, or "warbling"), going through zero twice per period of $\Delta f$.
- Even a small $\Delta f$ (a few Hz) makes speech unpleasant; it is a serious distortion, not a simple attenuation.

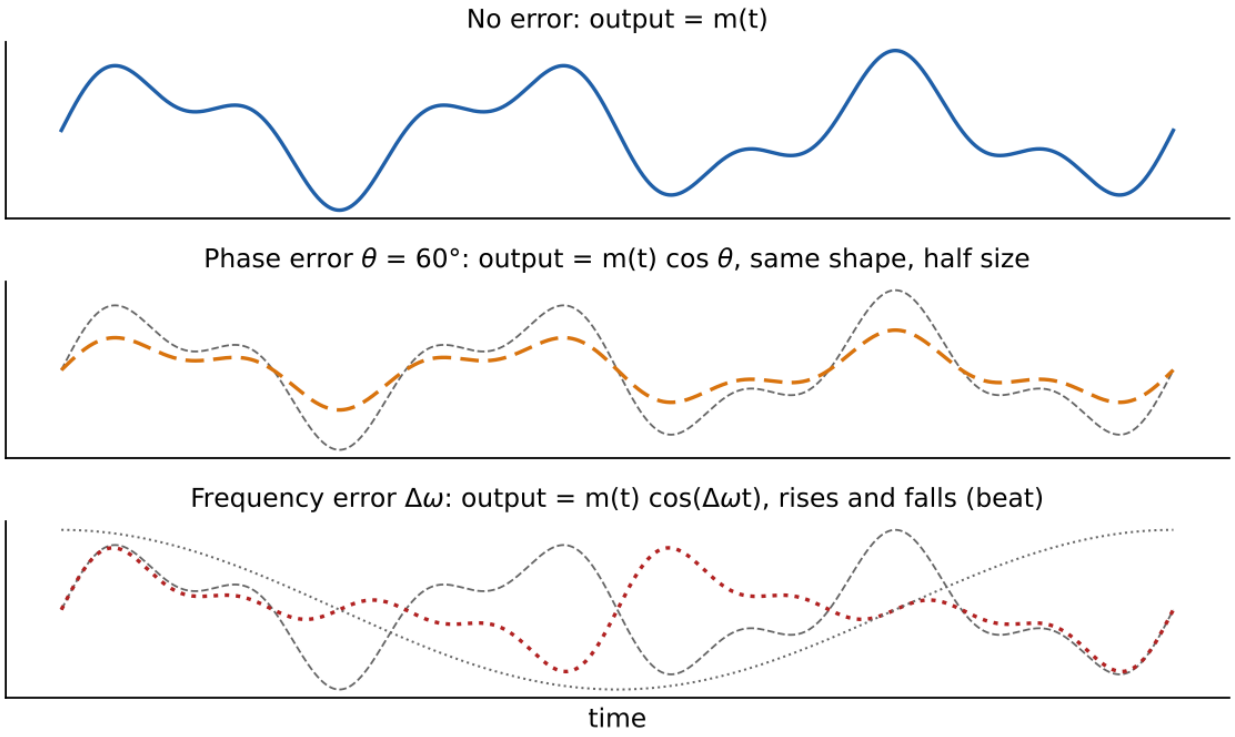



**Figure 3:** Effect of phase and frequency errors on the detected output

**Reading the figure.** The top trace is the correct output $m(t)$. In the middle, a phase error of 60° gives the same shape but only half as tall ($\cos 60° = 0.5$); the grey dashed curve is the correct output for comparison. At the bottom, a frequency error makes the output swell, shrink to zero and change sign as time goes on, following the dotted $\cos(\Delta\omega t)$ curve.


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**Practical Note:** to avoid both errors, the receiver must lock its oscillator to the incoming carrier: send a **pilot** carrier, use a **PLL**, or use a **Costas loop** (part c). Ordinary AM (DSB with a large carrier) avoids the problem by using an envelope detector, which needs no local carrier.

# 3. Notes and Remarks

For SSB the effect differs: a phase error only shifts the phase of every output component (tolerable for speech), and a frequency error shifts every output frequency by $\Delta f$ (a voice sounds like "Donald Duck").

**2.(c) With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver. Marks: 8**

# 1. Theory

## 1. The problem it solves (Core Idea)

From part (b), a DSB-SC receiver needs a local carrier with exactly the right frequency and phase. But a DSB-SC signal *contains no carrier* to copy. The **Costas receiver** (Costas loop) creates the carrier itself: it detects its own phase error and corrects it automatically, like a PLL.

## 2. The trick: two branches, I and Q

It demodulates the signal twice, with two local carriers 90° apart. The **in-phase (I)** branch gives the message times $\cos\theta$; the **quadrature (Q)** branch gives the message times $\sin\theta$. When $\theta = 0$ the Q branch is silent. So "Q is not zero" means "there is a phase error", and the loop uses this to steer the oscillator.

# 2. Answer

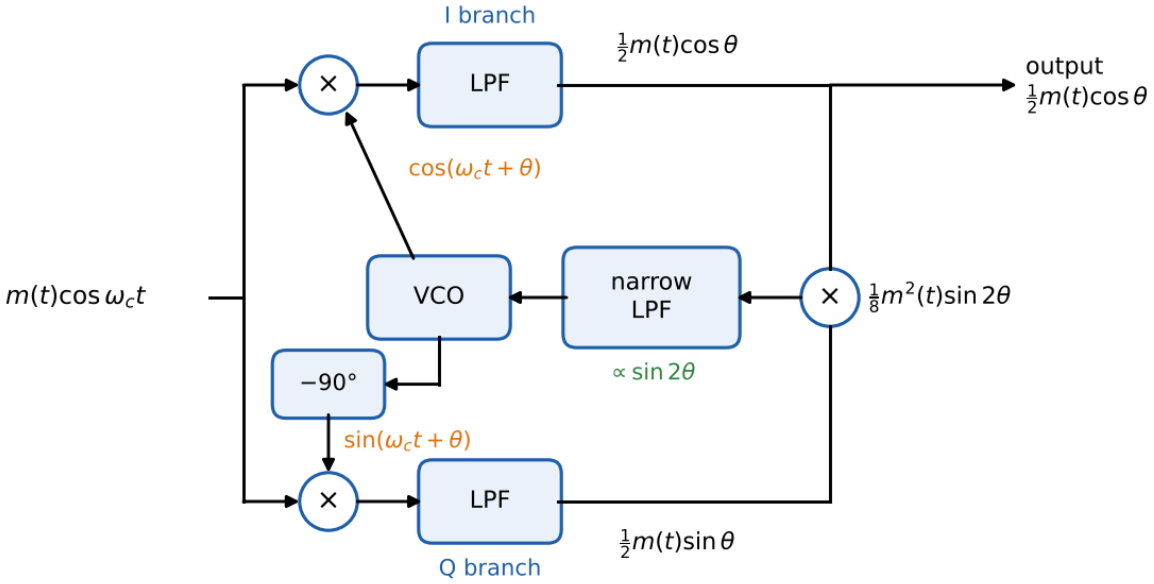



**Figure 4:** Costas receiver for DSB-SC

**Reading the figure.** The incoming DSB-SC signal $m(t)\cos\omega_c t$ splits into two branches. The VCO gives $\cos(\omega_c t+\theta)$ to the upper (I) multiplier and, through a $-90°$ shifter, $\sin(\omega_c t+\theta)$ to the lower (Q) multiplier. Each product goes through an LPF, giving $\frac{1}{2}m(t)\cos\theta$ (I) and $\frac{1}{2}m(t)\sin\theta$ (Q). A third multiplier on the right forms $\frac{1}{8}m^2(t)\sin 2\theta$, a narrow LPF smooths it, and the result (proportional to $\sin 2\theta$) controls the VCO. The output is taken from the I branch.

**Working principle:**

1. **I branch:** use $\cos A\cos B = \frac{1}{2}[\cos(B-A)+\cos(A+B)]$ with $A=\omega_c t$, $B=\omega_c t+\theta$, so $B-A=\theta$ and $A+B=2\omega_c t+\theta$:

$$m(t)\cos\omega_c t\cdot\cos(\omega_c t+\theta)=\tfrac{1}{2}m(t)[\cos\theta+\cos(2\omega_c t+\theta)]\xrightarrow{\text{LPF}}\tfrac{1}{2}m(t)\cos\theta$$

The LPF removes the term at $2\omega_c$.

2. **Q branch:** use $\cos A\sin B=\frac{1}{2}[\sin(B-A)+\sin(A+B)]$ with the same $A$ and $B$:

$$m(t)\cos\omega_c t\cdot\sin(\omega_c t+\theta)=\tfrac{1}{2}m(t)[\sin\theta+\sin(2\omega_c t+\theta)]\xrightarrow{\text{LPF}}\tfrac{1}{2}m(t)\sin\theta$$

3. **Phase detector (third multiplier):** multiply the two LPF outputs and use $\sin\theta\cos\theta=\frac{1}{2}\sin 2\theta$:

$$\tfrac{1}{2}m(t)\cos\theta\cdot\tfrac{1}{2}m(t)\sin\theta=\tfrac{1}{4}m^2(t)\sin\theta\cos\theta=\tfrac{1}{8}m^2(t)\sin 2\theta$$

4. **Narrow LPF:** averages $m^2(t)$ to its mean value $\overline{m^2}$, which is always positive. The control voltage is $\frac{1}{8}\overline{m^2}\sin 2\theta$. For a small angle $\sin x\approx x$, so $\sin 2\theta\approx 2\theta$ and the control voltage is $\approx\frac{1}{8}\overline{m^2}\cdot 2\theta=\frac{1}{4}\overline{m^2}\,\theta$: proportional to the phase error, with the right sign.
5. **VCO correction:** this voltage pushes the VCO frequency and phase so as to reduce $\theta$. The loop settles at $\theta\approx 0$.
6. **Output:** with $\theta\approx 0$, the I branch gives $\frac{1}{2}m(t)$, the demodulated message; the Q branch gives nearly zero.

**Key Points:**

- No pilot carrier needs to be transmitted, so all the transmitted power goes to the message.
- The control signal depends on $m^2(t)$, so the sign of $m(t)$ does not matter: the loop works whether the message is positive or negative.
- The loop also corrects a small frequency error, because a frequency error looks like a phase that keeps growing.

**Drawbacks:** the loop can also lock at $\theta=180°$ ($\sin 2\theta=0$ there too), giving $-\frac{1}{2}m(t)$: a sign inversion, harmless for audio. It needs time to lock, and loses lock when the message is silent for long.

# 3. Keyword Explanation

## 1. VCO (voltage-controlled oscillator)

An oscillator whose output frequency changes in proportion to an input control voltage.

## 2. DSB-SC

Double-sideband suppressed-carrier AM: $m(t)\cos\omega_c t$, with no separate carrier term, so it must be detected coherently.

**2.(d) Find the condition of sideband shaping filter for VSB modulation. Marks: 6**

# 1. Theory

## 1. Why VSB? (Core Idea)

DSB needs bandwidth $2B$. SSB needs only $B$, but cutting off one sideband exactly at $f_c$ needs an ideal, impossibly sharp filter, and it is hopeless for signals with important low frequencies (TV video). **Vestigial sideband (VSB)** is the compromise: keep one sideband fully and a small part (a **vestige**) of the other, using a filter with a gentle slope around $f_c$. Bandwidth is a little more than $B$.

## 2. What can go wrong?

The gentle filter changes the low frequencies of both sidebands. When the receiver demodulates, the two sidebands fold on top of each other. The filter must be shaped so that, after folding, the pieces add back to a flat response. That is the condition we need to find.

## 3. Two facts about filters that the answer uses

- **A filter multiplies the spectrum.** If a signal with spectrum $X(f)$ goes through a filter with transfer function $H(f)$, the output spectrum is $Y(f) = X(f)\,H(f)$. Each frequency is simply scaled by the filter's value at that frequency.
- **A real filter has a mirror-image response.** A filter built from real parts has a real impulse response $h(t)$. Then

$$H(-f) = \int_{-\infty}^{\infty} h(t)\, e^{+j2\pi ft}\, dt = \left[\int_{-\infty}^{\infty} h(t)\, e^{-j2\pi ft}\, dt\right]^{*} = H^{*}(f)$$

  The middle step holds because $h(t)$ is real, so taking the complex conjugate only flips the sign of $j$. In words: the response at $-f$ is the conjugate of the response at $+f$. For a filter drawn with a real (zero-phase) response, as in the figures here, this just means $H(-f) = H(f)$: the part of $H_i$ near $-f_c$ is the mirror image of the part near $+f_c$.

# 2. Answer

Below, the steps marked **Basic** build tools from first principles; a student who already knows them can skip straight to the result of that step. The notation follows Lathi (Sec. 4.5, Eqs. 4.23 to 4.26).

## Step 1: The VSB system

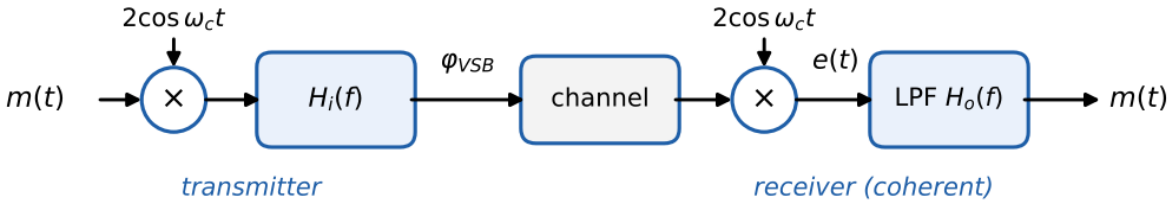



**Figure 5:** VSB transmitter and receiver (reused from the 2020-21 paper)

**Reading the figure.** Transmitter: the message $m(t)$ is multiplied by $2\cos\omega_c t$ (this makes DSB-SC), and the DSB-SC signal goes through the VSB shaping filter $H_i(f)$, giving $\varphi_{VSB}(t)$, which is sent through the channel. Receiver: $\varphi_{VSB}(t)$ is multiplied by the same carrier $2\cos\omega_c t$, giving $e(t)$, which goes through a low-pass filter $H_o(f)$.

The carrier is written as $2\cos\omega_c t$ (Lathi's choice) at both ends. The factor 2 makes the factors of $\frac{1}{2}$ in the steps below cancel, so the result is neater; it does not change the condition.

We must find what $H_i(f)$ has to satisfy so that the receiver's output is $m(t)$ without distortion.

## Step 2 (Basic): Multiplying by a cosine shifts a spectrum

Start from the definition of the Fourier transform, $X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$. Take the transform of $x(t)\, e^{j2\pi f_0 t}$:

$$\int_{-\infty}^{\infty} x(t)\, e^{j2\pi f_0 t}\, e^{-j2\pi ft}\, dt = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi (f-f_0)t}\, dt = X(f - f_0)$$

The two exponentials were joined into one ($e^a e^b = e^{a+b}$), and the last integral is the definition of $X$ with $f$ replaced by $f - f_0$. So

$$x(t)\, e^{j2\pi f_0 t} \Longleftrightarrow X(f - f_0)$$

Multiplying by $e^{j2\pi f_0 t}$ moves the whole spectrum to the right by $f_0$. In the same way, multiplying by $e^{-j2\pi f_0 t}$ moves it to the left: $X(f + f_0)$.

Now use Euler's formula, $e^{j\theta} + e^{-j\theta} = 2\cos\theta$, with $\theta = 2\pi f_c t = \omega_c t$:

$$2\cos\omega_c t = e^{j2\pi f_c t} + e^{-j2\pi f_c t}$$

So multiplying by $2\cos\omega_c t$ is the sum of one right shift and one left shift:

$$x(t)\cdot 2\cos\omega_c t \Longleftrightarrow X(f-f_c)+X(f+f_c)$$

This one result is used twice: once at the transmitter (Step 3) and once at the receiver (Step 4).

## Step 3: Spectrum of the VSB signal

**The DSB-SC signal.** Apply Step 2 with $x(t)=m(t)$:

$$2m(t)\cos\omega_c t \Longleftrightarrow M(f-f_c)+M(f+f_c)$$

The message spectrum appears twice, centred at $+f_c$ and at $-f_c$.

**The shaping filter.** The DSB-SC signal passes through $H_i(f)$, and a filter multiplies the spectrum (Theory 3):

$$\Phi_{VSB}(f)=\left[M(f+f_c)+M(f-f_c)\right]H_i(f) \qquad \text{(Lathi, Eq. 4.23)}$$

## Step 4: Demodulation: multiply by $2\cos\omega_c t$ again

The receiver forms $e(t)=2\varphi_{VSB}(t)\cos\omega_c t$. Apply Step 2 again, now with $x(t)=\varphi_{VSB}(t)$:

$$E(f)=\Phi_{VSB}(f+f_c)+\Phi_{VSB}(f-f_c)$$

Work out each shifted spectrum by writing $(f+f_c)$, and then $(f-f_c)$, in place of every $f$ in the result of Step 3.

**Left shift**, $f\to f+f_c$:

$$\Phi_{VSB}(f+f_c)=\left[M(f+f_c+f_c)+M(f+f_c-f_c)\right]H_i(f+f_c)=\left[M(f+2f_c)+M(f)\right]H_i(f+f_c)$$

**Right shift**, $f\to f-f_c$:

$$\Phi_{VSB}(f-f_c)=\left[M(f-f_c+f_c)+M(f-f_c-f_c)\right]H_i(f-f_c)=\left[M(f)+M(f-2f_c)\right]H_i(f-f_c)$$

**Add the two.** Multiply out the brackets:

$$E(f)=M(f+2f_c)H_i(f+f_c)+M(f)H_i(f+f_c)+M(f)H_i(f-f_c)+M(f-2f_c)H_i(f-f_c)$$

The second and third terms both contain $M(f)$, so take $M(f)$ out as a common factor:

$$E(f)=\underbrace{M(f)\left[H_i(f+f_c)+H_i(f-f_c)\right]}_{\text{near } f=0}+\underbrace{M(f+2f_c)H_i(f+f_c)+M(f-2f_c)H_i(f-f_c)}_{\text{near } f=\mp 2f_c}$$

## Step 5: The low-pass filter removes the copies at $\pm 2f_c$

$M(f)$ is non-zero only for $|f|\le B$. So $M(f-2f_c)$, which is $M$ moved to the right by $2f_c$, is non-zero only for

$$2f_c-B\le f\le 2f_c+B$$

Its lowest frequency, $2f_c-B$, is above $B$ whenever $f_c>B$, which is always true (the carrier is far above the highest message frequency). So this term lies completely outside the message band $|f|\le B$. In the same way, $M(f+2f_c)$ lies around $-2f_c$, completely below $-B$.


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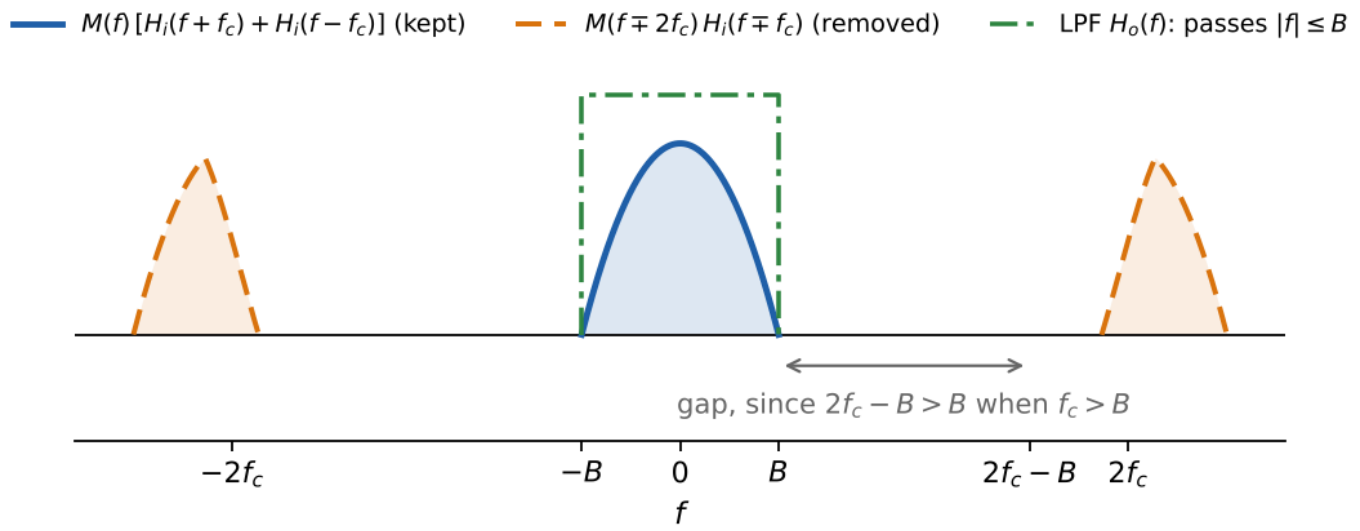



**Figure 6:** Spectrum after the receiver's multiplier

**Reading the figure.** The solid blue hump in the middle is the wanted term $M(f)[H_i(f+f_c)+H_i(f-f_c)]$, inside $|f| \le B$. The dashed orange humps are the copies around $\pm 2f_c$; each keeps the VSB shape (one sideband and a vestige). The dash-dot green box is the low-pass filter $H_o(f)$, which passes only $|f| \le B$. The grey arrow shows the empty gap between $B$ and $2f_c - B$, so the filter can keep the middle term and reject the copies.

The low-pass filter $H_o(f)$ passes $|f| \le B$ and rejects everything else, so its output spectrum is

$$E_o(f) = M(f)\,[H_i(f+f_c) + H_i(f-f_c)]\,H_o(f), \qquad |f| \le B$$

## Step 6: The condition for a distortionless output

We want the output to be the message itself, $E_o(f) = M(f)$. Comparing with Step 5, everything that multiplies $M(f)$ must equal 1:

$$[H_i(f+f_c) + H_i(f-f_c)]\,H_o(f) = 1$$

Divide by the bracket:

$$H_o(f) = \frac{1}{H_i(f+f_c) + H_i(f-f_c)}, \quad |f| \le B \qquad \text{(Lathi, Eq. 4.25)}$$

This is the general condition: whatever $H_i$ is, the receiver filter must undo the factor in the bracket.

The usual receiver uses a plain low-pass filter, $H_o(f) = 1$ for $|f| \le B$. Then the bracket alone must give the same gain to every message frequency. This is exactly the distortionless condition of 1(d): a constant gain over the whole band of the signal. So the condition on the shaping filter is

$$\boxed{H_i(f+f_c) + H_i(f-f_c) = \text{constant} \quad \text{for } |f| \le B}$$

Lathi takes the constant as 1 (Eq. 4.26), the **complementary VSB filter**.

## Step 7: What the condition means for the shape of $H_i(f)$

For $|f| \le B$, the term $H_i(f+f_c)$ reads the filter near $+f_c$, and $H_i(f-f_c)$ reads it near $-f_c$. To see the shape, rename the message frequency $f$ as $x$ (so $|x| \le B$, the distance from the carrier) and take the constant as 1:

$$H_i(f_c + x) + H_i(x - f_c) = 1$$

**Bring both terms to the side near $+f_c$.** By the mirror property of a real filter (Theory 3), $H_i(x - f_c) = H_i(-(f_c - x)) = H_i^*(f_c - x)$. For the real (zero-phase) filter drawn here, $H_i^* = H_i$. So the condition becomes

$$H_i(f_c + x) + H_i(f_c - x) = 1, \qquad |x| \le B$$


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**The value at the carrier.** Put $x = 0$: $2H_i(f_c) = 1$, so

$$H_i(f_c) = \tfrac{1}{2}$$

The filter passes exactly half of the carrier frequency.

**The symmetry.** Write the filter near $f_c$ as half plus a deviation, $H_i(f_c + x) = \frac{1}{2} + g(x)$. Then $H_i(f_c - x) = \frac{1}{2} + g(-x)$. Putting both into the condition:

$$\tfrac{1}{2} + g(x) + \tfrac{1}{2} + g(-x) = 1 \quad \Rightarrow \quad g(-x) = -g(x)$$

So $g$ is an odd function: the filter has **odd (complementary) symmetry about the point** $(f_c, \frac{1}{2})$. As far as the filter rises above $\frac{1}{2}$ at $f_c + x$, it falls below $\frac{1}{2}$ at $f_c - x$.

**Away from the carrier.** Outside the vestige the filter is fully on above $f_c$, $H_i(f_c + x) = 1$; then the condition gives $H_i(f_c - x) = 0$. So the filter passes one sideband completely and cuts the other one completely, except for the vestige near $f_c$.

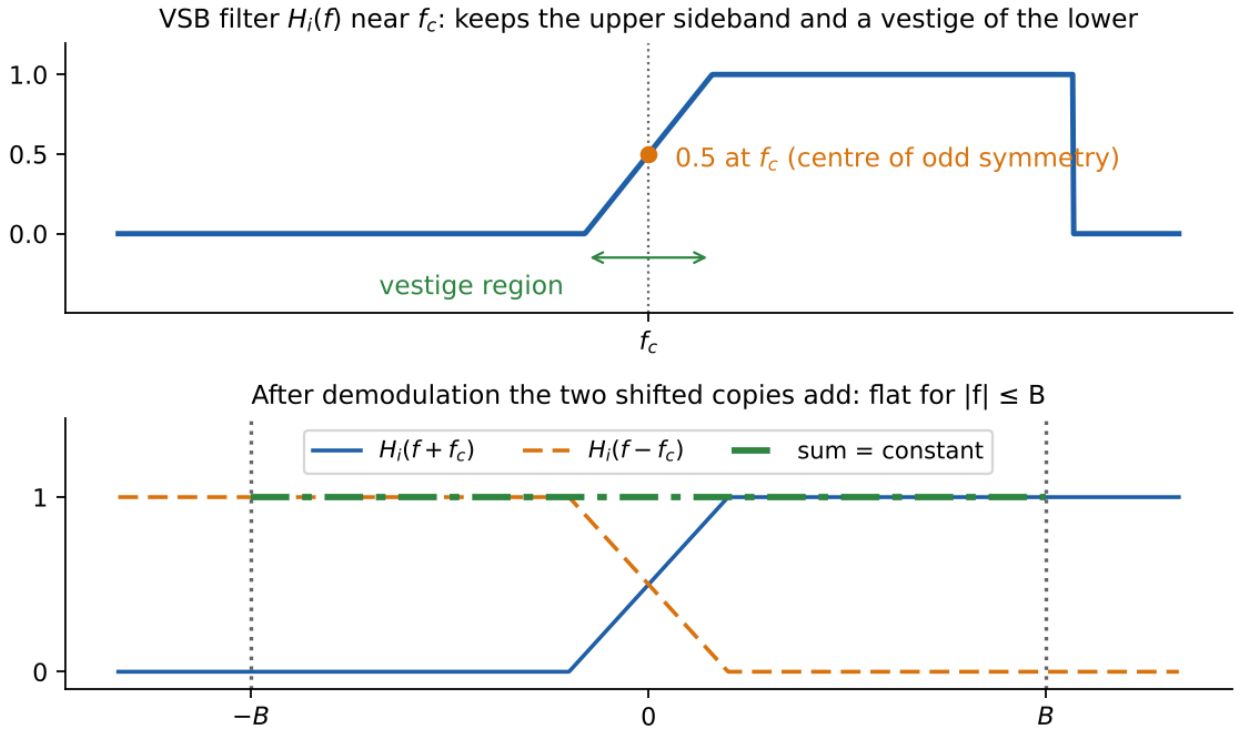



**Figure 7:** VSB filter shape and the complementary-symmetry condition

**Reading the figure.** The top plot shows the VSB filter near $f_c$: it rises from 0 to 1 across a small "vestige region", passing through 0.5 at $f_c$. The bottom plot shows the two shifted copies after demodulation. The solid blue one rises where the dashed orange one falls, so their sum (green, dash-dot) stays flat for $|f| \le B$.

**Interpretation:** the part of the wanted sideband that the gentle slope cut away is exactly filled in, after demodulation, by the vestige of the unwanted sideband. That is why the gentle, buildable filter still gives a perfect output.

**Steps to remember**

DSB-SC → shape with $H_i$ → multiply by $2\cos\omega_c t$ (two shifts by $\pm f_c$) → add and collect $M(f)$ → LPF removes the $\pm 2f_c$ copies → the bracket $H_i(f + f_c) + H_i(f - f_c)$ must be flat → half gain and odd symmetry at $f_c$.

**Practical Note:** TV broadcasting used VSB: a 4.2 MHz video signal is sent with a vestige of about 1.25 MHz of the lower sideband, in a 6 MHz channel.


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# 3. Final Concise Answer

**(a)** $I_t = I_c\sqrt{1+\mu^2/2}$: $\mu = \sqrt{2[(8.93/8)^2-1]} = 0.7015$, about 70% modulation; at $\mu = 0.8$, $I_t = 8\sqrt{1.32} \approx 9.19$ A.

**(b)** With local carrier $2\cos[(\omega_c+\Delta\omega)t+\theta]$ the output is $m(t)\cos(\Delta\omega t+\theta)$: a phase error scales the output by $\cos\theta$ (zero at 90°, fading if $\theta$ varies); a frequency error multiplies it by $\cos\Delta\omega t$ (beating). Hence the need for carrier synchronization.

**(c)** The Costas loop uses I and Q branches with a VCO; the LPF outputs are $\frac{1}{2}m\cos\theta$ and $\frac{1}{2}m\sin\theta$; their product $\frac{1}{8}m^2\sin 2\theta$, after a narrow LPF, drives the VCO to $\theta = 0$; the I branch then gives $\frac{1}{2}m(t)$.

**(d)** $\Phi_{VSB} = [M(f+f_c)+M(f-f_c)]H_i(f)$; multiplying by $2\cos\omega_c t$ gives $E(f) = M(f)[H_i(f+f_c)+H_i(f-f_c)]$ plus copies at $\pm 2f_c$, which the LPF removes. So $H_o = 1/[H_i(f+f_c)+H_i(f-f_c)]$, and with a plain LPF, $H_i(f+f_c)+H_i(f-f_c) =$ constant for $|f| \le B$: the filter is odd-symmetric about $f_c$, with half gain at $f_c$.


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3. (a) Explain NBFM generation process with necessary block diagrams and equations. Marks: 5

(b) Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling. Marks: 5

(c) A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz.

(i) Find the carrier swing of the FM signal.

(ii) What are the highest and lowest frequencies attained by the frequency modulated signal?

(iii) Calculate the modulation index for the wave. Marks: 5

(d) For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$. Marks: 10

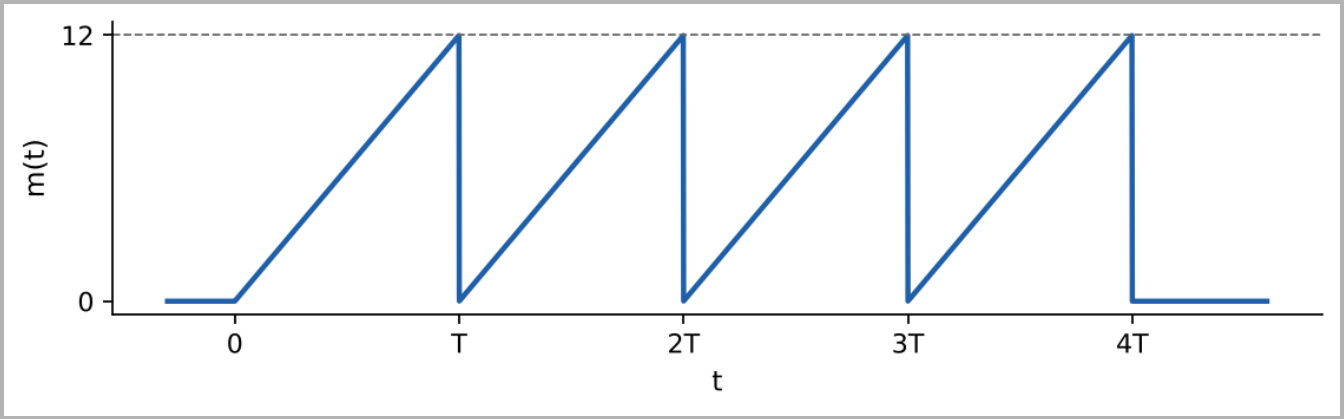



**Figure 8:** Fig. 1 of the paper (redrawn): the message m(t)

**3.(a) Explain NBFM generation process with necessary block diagrams and equations. Marks: 5**

# 1. Theory

## 1. What is FM, and what is "narrowband"? (Core Idea)

In **frequency modulation (FM)** the carrier keeps a constant amplitude, but its instantaneous frequency moves up and down with the message: $\omega_i(t) = \omega_c + k_f m(t)$. Since phase is the integral of frequency, the FM wave is

$$\varphi_{FM}(t) = A\cos\left[\omega_c t + k_f \int m(\tau)\, d\tau\right] = A\cos[\omega_c t + k_f a(t)]$$

where $a(t) = \int m(\tau)\, d\tau$. FM is called **narrowband (NBFM)** when the phase deviation stays small, $|k_f a(t)| \ll 1$ (modulation index $\beta \ll 1$). Its bandwidth is then only about $2B$, like AM.

## 2. Why is NBFM worth making?

Wideband FM is hard to make with a stable carrier. NBFM can be made from a crystal oscillator and simple blocks, and then widened by frequency multipliers (the indirect or Armstrong method). So NBFM is the first stage of a broadcast FM transmitter.

# 2. Answer

## Step 1: The FM wave

In FM the instantaneous frequency follows the message, $\omega_i(t) = \omega_c + k_f m(t)$. The angle of a carrier is the integral of its instantaneous frequency (because $\omega_i = d\theta/dt$), so

$$\theta(t) = \int_{-\infty}^{t} \omega_i(\tau)\, d\tau = \omega_c t + k_f \int_{-\infty}^{t} m(\tau)\, d\tau = \omega_c t + k_f a(t)$$

(the constant of integration is taken as zero). So $\varphi_{FM}(t) = A\cos\theta(t) = A\cos[\omega_c t + k_f a(t)]$.

## Step 2: Expand the cosine

Use $\cos(X+Y) = \cos X \cos Y - \sin X \sin Y$ with $X = \omega_c t$ and $Y = k_f a(t)$:

$$\varphi_{FM}(t) = A\cos\omega_c t\, \cos[k_f a(t)] - A\sin\omega_c t\, \sin[k_f a(t)]$$

## Step 3: Use the small angle of narrowband FM

The power series are $\cos Y = 1 - \frac{Y^2}{2!} + \ldots$ and $\sin Y = Y - \frac{Y^3}{3!} + \ldots$. For narrowband FM, $|Y| = |k_f a(t)| \ll 1$, so the squared and cubed terms are tiny and only the first term of each series is kept: $\cos[k_f a(t)] \approx 1$ and $\sin[k_f a(t)] \approx k_f a(t)$. Then

$$\boxed{\varphi_{NBFM}(t) \approx A\cos\omega_c t - A\, k_f\, a(t)\sin\omega_c t, \qquad a(t) = \int m(\tau)\, d\tau}$$

So NBFM is **the carrier minus a DSB-SC signal** of the integrated message on a carrier shifted by 90°.

## Step 4: Block diagram from the formula

The formula tells us exactly which blocks to use.

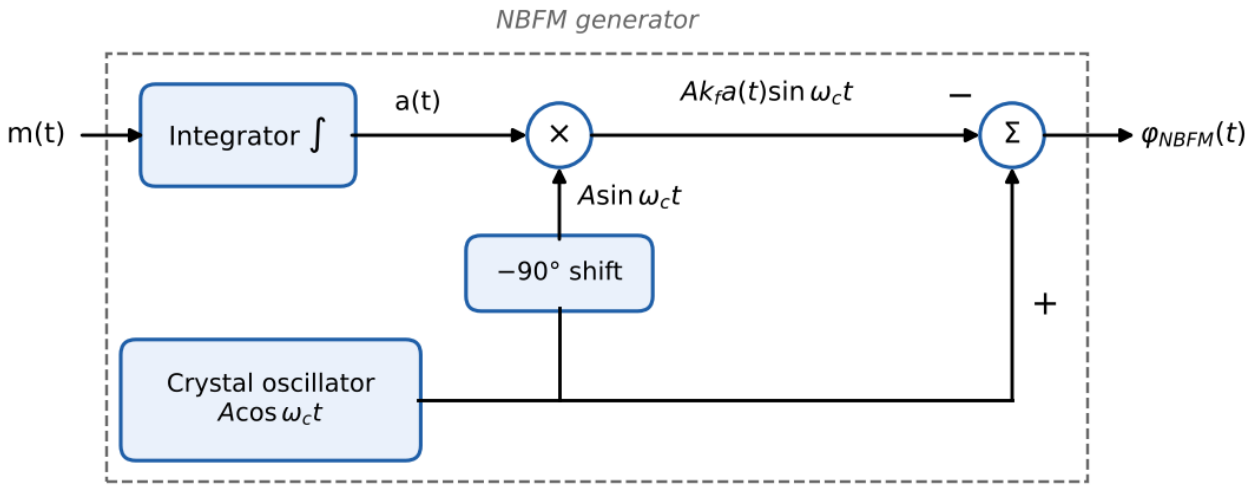



**Figure 9:** NBFM generator

### Generation process:

1. The message $m(t)$ is passed through an **integrator**, giving $a(t) = \int m\, d\tau$.
2. A stable **crystal oscillator** gives the carrier $A\cos\omega_c t$.
3. A $-90°$ **phase shifter** turns the carrier into $A\sin\omega_c t$.
4. A **balanced modulator (multiplier)** forms $Ak_f a(t)\sin\omega_c t$, a DSB-SC signal.
5. A **summer** subtracts this from the carrier: $A\cos\omega_c t - Ak_f a(t)\sin\omega_c t = \varphi_{NBFM}(t)$.

### Key Points:

- Without the integrator the same circuit gives **NBPM** (phase modulation). Integrating first turns PM into FM.
- Compare with AM, $A\cos\omega_c t + m(t)\cos\omega_c t$: NBFM has the sideband term in **quadrature** (on $\sin\omega_c t$) instead of in phase. That is why its amplitude stays almost constant.

- Bandwidth $\approx 2B$.

**Practical Note:** in the Armstrong (indirect) transmitter, the NBFM output (with $\beta$ well below 0.5) is passed through frequency multipliers and a mixer to reach the wanted carrier and deviation, giving WBFM.

**3.(b) Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling. Marks: 5**

# 1. Theory

## 1. What is the envelope? (Core Idea)

An AM wave $[A + m(t)]\cos\omega_c t$ is a fast carrier whose peaks trace out the slow curve $A + m(t)$. That curve is the **envelope**, and it carries the message. An **envelope detector** draws a line through the positive peaks of the carrier, giving the envelope.

## 2. Why "linear diode" detection?

The diode works with a large input signal, where it acts as a simple on/off switch (conducting only on positive peaks). The output is then directly (linearly) proportional to the envelope. With a weak input the diode works in its curved (square-law) region instead.

# 2. Answer

The linear diode detector (envelope detector) is the simplest AM demodulator. It consists of a diode in series with a parallel **RC** load.

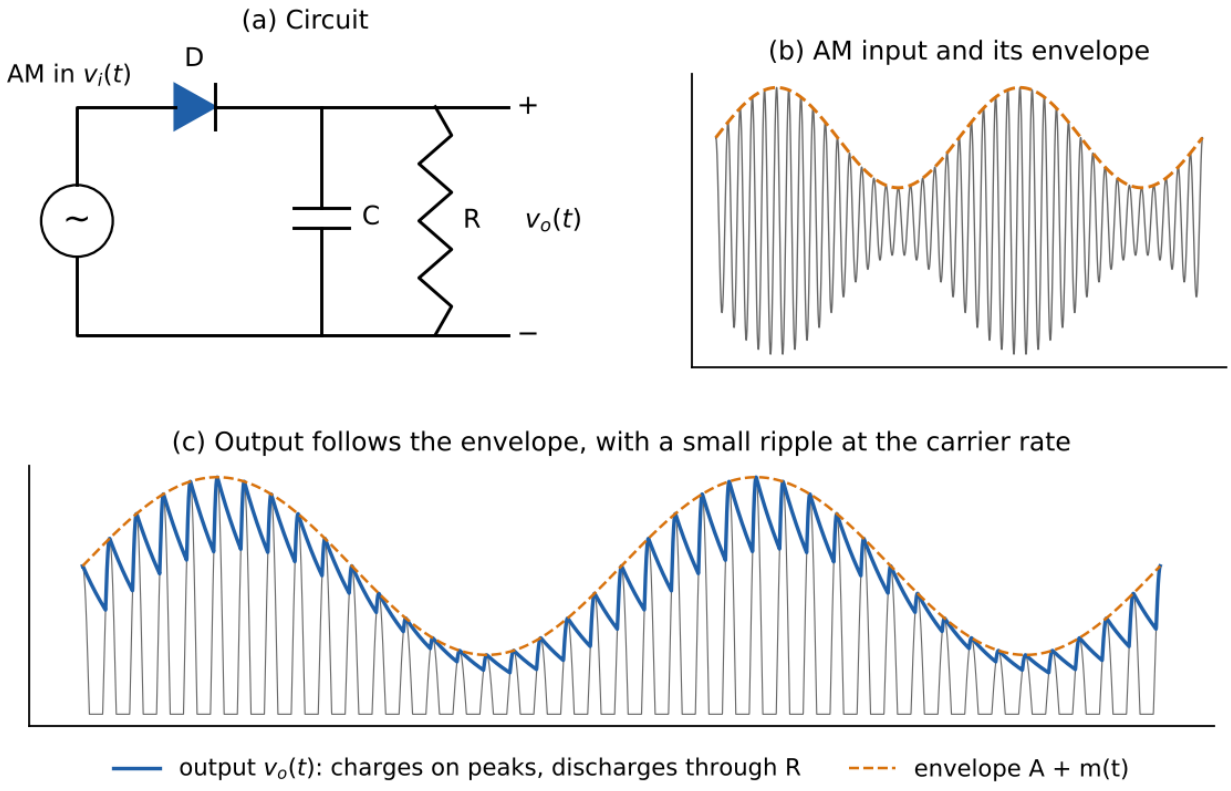



**Figure 10:** Envelope detector: circuit, input and output

**Reading the figure.** (a) The circuit: the AM source feeds the diode D; the capacitor C and resistor R are in parallel across the output $v_o(t)$. (b) The input AM wave (grey) with its envelope (orange dashed). (c) The output (solid blue): it jumps up to each positive peak and then slopes down slowly until the next peak, so it follows the envelope with a small ripple.

**Working:**

1. **Charging:** on each positive half-cycle, when the input is higher than the capacitor voltage, the diode conducts and C charges quickly to the peak value of the carrier.
2. **Discharging:** when the input falls below the capacitor voltage, the diode switches off and C discharges slowly through R.
3. At the next peak the diode conducts again and tops up C. So the capacitor voltage follows the peaks, that is, the envelope $A + m(t)$, with a small ripple at the carrier frequency.
4. A DC-blocking capacitor removes $A$, and a small RC filter smooths the ripple, leaving $m(t)$.

**Choosing the time constant RC:**

$$\boxed{\frac{1}{\omega_c} \ll RC < \frac{1}{2\pi B}}$$

where:

- $\omega_c$ = carrier frequency (rad/s), $B$ = highest message frequency (Hz).
- **RC too small:** C discharges almost fully between peaks; the output has a large ripple (follows the carrier).
- **RC too large:** C cannot discharge fast enough when the envelope falls; the output misses the falling parts (**diagonal clipping**).

**Condition for correct detection:** the envelope must never go negative: $A + m(t) \geq 0$, i.e. $\mu \leq 1$. With over-modulation the detector gives a distorted output.

**Advantages:** very simple and cheap (one diode, one R, one C); no local carrier needed (non-coherent). That is why every AM broadcast radio uses it.

**Disadvantages:** works only for AM with a carrier and $\mu \leq 1$; not for DSB-SC or SSB; ripple and diagonal clipping if RC is badly chosen.

# 3. Notes and Remarks

The paper's word "leveling" is read as "labelling": the diagrams above label the diode D, capacitor C, load R, input $v_i(t)$, output $v_o(t)$ and the envelope.

**3.(c) A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz. (i) Find the carrier swing of the FM signal. (ii) What are the highest and lowest frequencies attained by the frequency modulated signal? (iii) Calculate the modulation index for the wave. Marks: 5**

# 1. Theory

## 1. Frequency deviation (Core Idea)

In FM the instantaneous frequency moves above and below the carrier. The largest distance it moves from $f_c$ is the **frequency deviation** $\Delta f$. So the frequency stays between $f_c - \Delta f$ and $f_c + \Delta f$.

## 2. Carrier swing and modulation index

- **Carrier swing** is the total distance from the lowest to the highest frequency: $2\Delta f$.
- **Modulation index** $\beta = \Delta f / f_m$: how many times bigger the deviation is than the message frequency. It decides whether the FM is narrowband ($\beta \ll 1$) or wideband ($\beta \gg 1$).

# 2. Answer

**Given:**

- Carrier $f_c = 93.2$ MHz
- Modulating frequency $f_m = 5$ kHz
- Frequency deviation $\Delta f = 40$ kHz


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### Step 1: Carrier swing

$$\text{Carrier swing} = 2\Delta f = 2 \times 40 \text{ kHz} = 80 \text{ kHz}$$

$$\boxed{\text{Carrier swing} = 80 \text{ kHz}}$$

### Step 2: Highest and lowest frequencies

$$f_{max} = f_c + \Delta f = 93.2 \text{ MHz} + 0.04 \text{ MHz} = 93.24 \text{ MHz}$$

$$f_{min} = f_c - \Delta f = 93.2 \text{ MHz} - 0.04 \text{ MHz} = 93.16 \text{ MHz}$$

$$\boxed{f_{max} = 93.24 \text{ MHz}, \quad f_{min} = 93.16 \text{ MHz}}$$

### Step 3: Modulation index

$$\beta = \frac{\Delta f}{f_m} = \frac{40 \text{ kHz}}{5 \text{ kHz}} = 8$$

$$\boxed{\beta = 8}$$

Since $\beta \gg 1$ this is wideband FM.
**Final Answer:** carrier swing = 80 kHz; frequency range 93.16 MHz to 93.24 MHz; $\beta = 8$.

## 3. Notes and Remarks

By Carson's rule the bandwidth is $B_T \approx 2(\Delta f + f_m) = 2(40+5) = 90$ kHz. Note that the carrier swing (80 kHz) is not the bandwidth: the sidebands reach a little beyond the frequency limits.

**3.(d) For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$. Marks: 10**

## 1. Theory

### 1. The AM wave (Core Idea)

Full AM (DSB with carrier) is

$$\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t$$

It is a fast carrier whose **envelope** (the curve through its peaks) is $A + m(t)$, and its mirror image $-[A + m(t)]$ at the bottom. To sketch an AM wave you only need to draw the envelope and fill it with a carrier.

### 2. Modulation index

$$\mu = \frac{m_p}{A}$$

where $m_p$ is the peak value of $m(t)$. So for a given $m(t)$, choosing $\mu$ means choosing the carrier amplitude $A = m_p/\mu$. A small $\mu$ means a big carrier compared with the message.

### 3. What to watch

The envelope detector of part (b) works only while the envelope $A + m(t)$ stays $\geq 0$. If it goes negative, the carrier flips phase and the envelope no longer equals $A + m(t)$: this is **over-modulation**.


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# 2. Answer

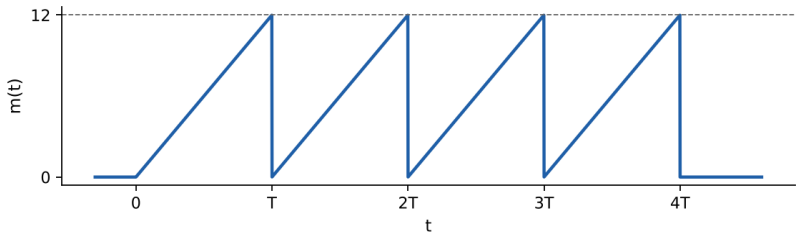



**Figure 11:** Repeated from the question: Fig. 1 of the paper (redrawn): the message m(t)

**Given:** $m(t)$ is a sawtooth (Fig. 1): it rises linearly from 0 to 12 in each period $T$ and drops back to 0, for $0 \le t < 4T$. So $m_{min} = 0$ and $m_p = 12$.
**Carrier amplitude for each $\mu$:**

$$A = \frac{m_p}{\mu} = \frac{12}{\mu}$$

| $\mu$ | $A = 12/\mu$ | Envelope $A + m(t)$ | Lowest envelope | Highest envelope |
|---|---|---|---|---|
| 0.5 | 24 | $24 + m(t)$ | 24 | 36 |
| 1 | 12 | $12 + m(t)$ | 12 | 24 |
| 1.5 | 8 | $8 + m(t)$ | 8 | 20 |

**Table 2:** Carrier amplitude and envelope limits for each modulation index

**How to sketch:** for each $\mu$, draw the upper envelope $A + m(t)$: a sawtooth that starts at $A$, rises to $A + 12$ at the end of each period and drops back to $A$. Draw its mirror image $-[A + m(t)]$ below the axis. Fill the space between them with a carrier that touches both envelopes.

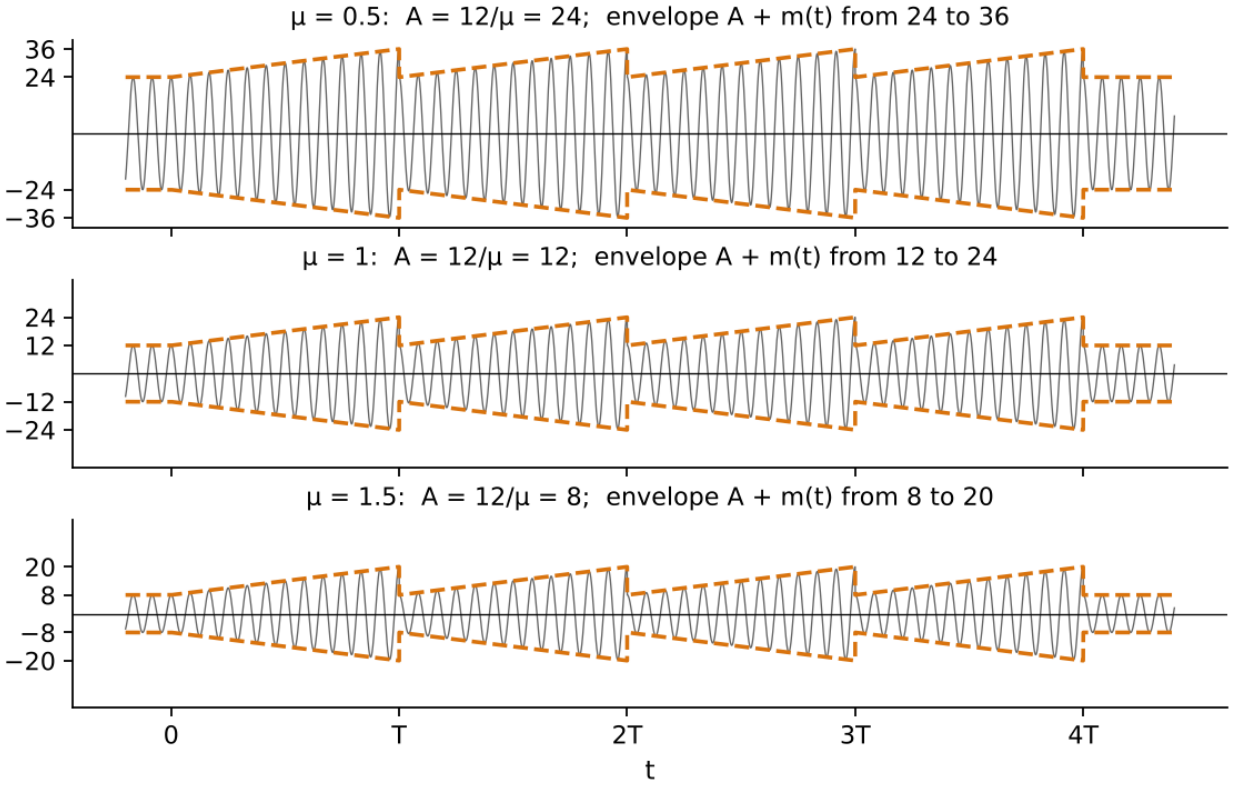



**Figure 12:** Sketch of the AM signal for mu = 0.5, 1 and 1.5


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**Reading the figure.** Each panel shows the carrier (grey) inside its envelope (orange dashed). Top, $\mu = 0.5$: the envelope swings between 24 and 36, so the variation is small compared with the carrier. Middle, $\mu = 1$: between 12 and 24. Bottom, $\mu = 1.5$: between 8 and 20; the carrier is now small compared with the message, but the envelope still stays above zero.

**Observations:**

- As $\mu$ grows, the carrier $A$ gets smaller, so the message takes a bigger share of the wave (better power efficiency).
- Because this $m(t)$ is never negative, $A + m(t) \geq A > 0$ for all three values. The envelope never touches zero, and the message can be recovered by an envelope detector in every case, even at $\mu = 1.5$.

# 3. Notes and Remarks

**If the dc part of $m(t)$ is removed first** (the usual practice, since a dc level in the message only adds to the carrier), the message becomes $m_0(t) = m(t) - 6$, a sawtooth from $-6$ to $+6$, so $m_p = 6$ and $A = 6/\mu$: $A = 12, 6, 4$ for $\mu = 0.5, 1, 1.5$. The envelope $A + m_0(t)$ then runs from 6 to 18, from 0 to 12, and from $-2$ to 10. For $\mu = 1$ it just touches zero, and for $\mu = 1.5$ it goes negative: **over-modulation**, with phase reversal of the carrier and a distorted output from an envelope detector.

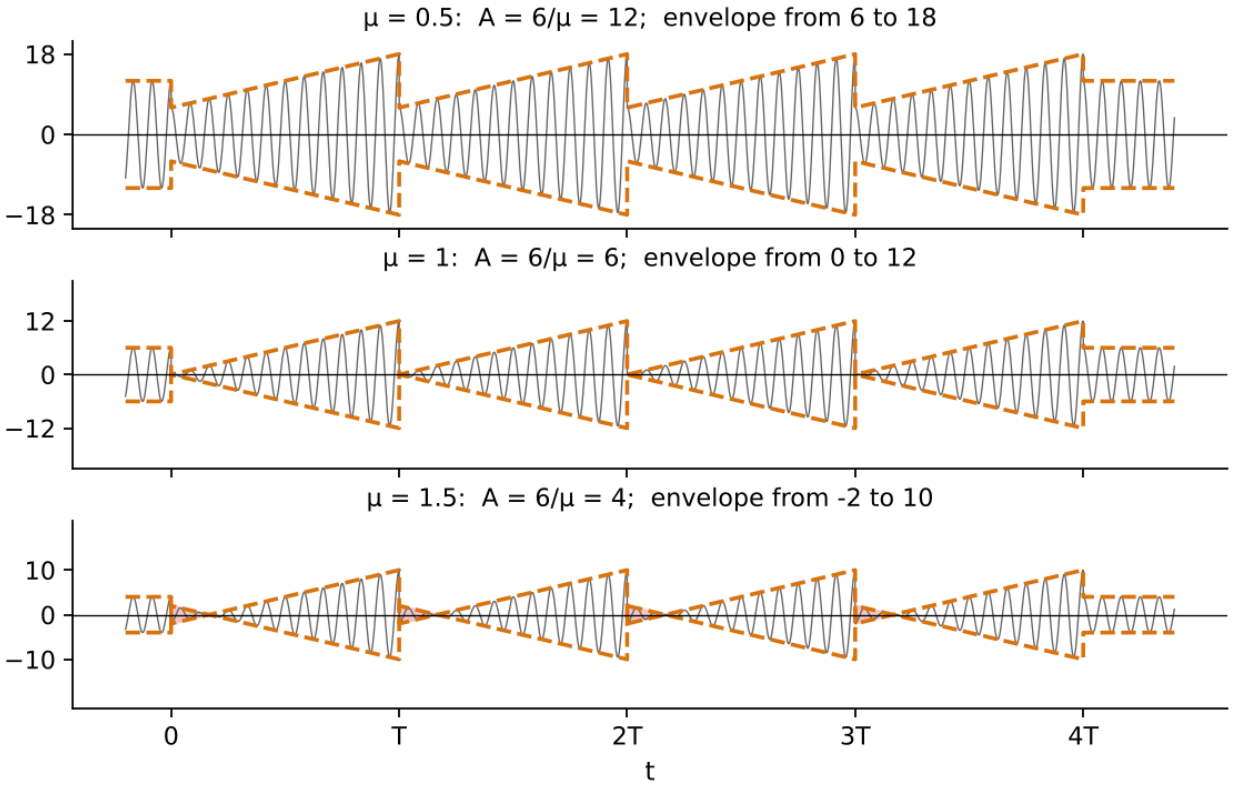



**Figure 13:** Same sketch with the dc part of m(t) removed

**Reading the figure.** The shaded red parts in the bottom panel ($\mu = 1.5$) are where the envelope $A + m_0(t)$ is below zero; there the carrier is turned upside down (phase reversal), so its peaks no longer follow the message.

**Exam tip**

Say which convention you use. With $\mu = m_p/A$ and the sawtooth taken as drawn (0 to 12), no over-modulation occurs; with the dc removed ($\pm 6$), $\mu = 1.5$ gives over-modulation.

# 4. Final Concise Answer

(a) $\varphi_{NBFM} \approx A\cos\omega_c t - Ak_f a(t)\sin\omega_c t$ with $a(t) = \int m\,dt$: integrator $\rightarrow$ balanced modulator with

$A \sin \omega_c t$ (carrier shifted by $-90°$) $\rightarrow$ subtract from the carrier.

**(b)** Diode + parallel RC: C charges to each carrier peak and discharges slowly through R, so $v_o$ follows the envelope $A + m(t)$; needs $1/\omega_c \ll RC < 1/(2\pi B)$ and $\mu \leq 1$.

**(c)** Carrier swing $= 2\Delta f = 80$ kHz; $f_{max} = 93.24$ MHz, $f_{min} = 93.16$ MHz; $\beta = 40/5 = 8$.

**(d)** $A = 12/\mu = 24, 12, 8$; envelope $A + m(t)$ is a sawtooth from $A$ to $A + 12$ (24–36, 12–24, 8–20) and its mirror image; it never reaches zero for this non-negative $m(t)$ (with the dc removed, $\mu = 1.5$ would over-modulate).


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4. (a) What is PCM? Explain PCM system. Marks: 5
   (b) "A proper valued LPF filter can be used to reconstruct the sampled signal"-justify the statement with the proper mathematical explanations. Marks: 5
   (c) Write short note on: Companding, Information rate, and Entropy of a source. Marks: 5
   (d) Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them. Marks: 5
   (e) An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find-
      (i) Peak signal power to peak quantization noise power.
      (ii) System bandwidth. Marks: 5

**4.(a) What is PCM? Explain PCM system. Marks: 5**

# 1. Theory

## 1. Why turn an analog signal into numbers? (Core Idea)

Noise adds to an analog signal and can never be fully removed. A digital signal has only two values, 0 and 1. As long as noise is not big enough to turn a 0 into a 1, a repeater can **regenerate** a perfectly clean signal. PCM is the standard way of turning an analog signal (such as voice) into bits.

## 2. The three steps

To describe a signal with numbers we must (1) look at it only at regular instants (**sampling**), (2) round each value to one of a fixed set of levels (**quantization**), and (3) write the level as a binary number (**encoding**).

# 2. Answer

**Definition:** **Pulse code modulation (PCM)** is a method of converting an analog signal into a digital (binary) signal by sampling it, quantizing each sample to one of $L$ levels, and encoding each level as an $n$-bit binary code word ($L = 2^n$).

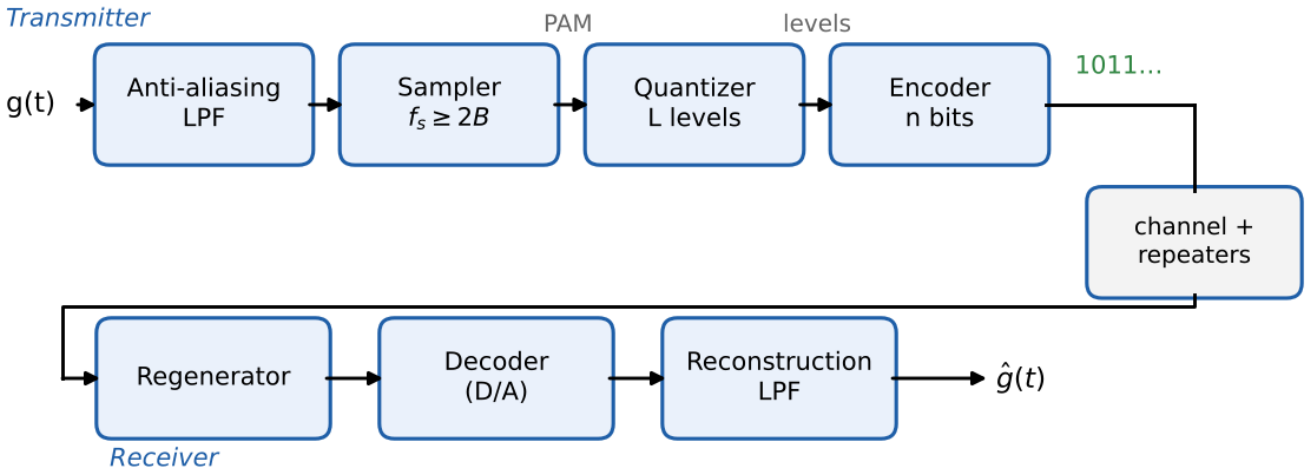



**Figure 14:** PCM system: transmitter, channel and receiver

**Reading the figure.** Top row (transmitter), left to right: the analog signal $g(t)$ passes the anti-aliasing LPF, the sampler (output: PAM samples), the quantizer (output: quantized levels) and the encoder (output: a bit stream such as 1011…). The bits travel through the channel and its regenerative repeaters. Bottom row (receiver): the regenerator cleans up the pulses, the decoder turns code words back into levels, and the reconstruction LPF gives the analog output $\hat{g}(t)$.

**Transmitter:**

1. **Anti-aliasing LPF:** limits the message to bandwidth $B$ so that sampling causes no aliasing.
2. **Sampler:** takes samples at $f_s \geq 2B$ (Nyquist rate), giving PAM pulses.
3. **Quantizer:** rounds each sample to the nearest of $L$ levels. The rounding error is the **quantization noise**, with power $\Delta^2/12$ for step size $\Delta = 2m_p/L$.
4. **Encoder:** converts each level into an $n = \log_2 L$ bit code word. The output bit rate is $R_b = nf_s$.

**Channel:** the bits are sent as pulses (a line code). **Regenerative repeaters** along the way detect each bit and send a fresh clean pulse, so noise does not build up.

**Receiver:**

1. **Regenerator:** decides 0 or 1 for each pulse.
2. **Decoder (D/A):** turns each code word back into its quantized level (PAM samples).
3. **Reconstruction LPF:** smooths the samples back into the analog signal.

**Key Points:**

- SQNR $= 3L^2\,\overline{m^2}/m_p^2$; in dB about $6.02n +$ constant: each extra bit adds 6 dB.
- Minimum bandwidth $B_T = R_b/2 = nB$ (at Nyquist sampling).
- **Advantages:** noise immunity, regeneration, easy encryption and multiplexing (TDM), uses cheap digital hardware. **Disadvantages:** needs much more bandwidth than the analog signal; quantization noise is always present.

**4.(b) "A proper valued LPF filter can be used to reconstruct the sampled signal"-justify the statement with the proper mathematical explanations. Marks: 5**

# 1. Theory

## 1. What does sampling do to the spectrum? (Core Idea)

Sampling multiplies the signal by a train of impulses one $T_s$ apart. In frequency this makes **copies** of the original spectrum $G(f)$, repeated every $f_s = 1/T_s$. If the copies do not overlap, the original copy (centred at 0) is still there, untouched. A low-pass filter can then keep it and throw the others away. That is the whole idea; the maths below makes it exact.

## 2. What "proper valued" means

The filter must have the right **cutoff** (passing the original copy, stopping all the others) and the right **gain** ($T_s$, to undo the $1/T_s$ scaling that sampling introduced).

# 2. Answer

## Step 1: Write sampling as a multiplication

Let $g(t)$ be band-limited to $B$ Hz: its spectrum $G(f) = 0$ for $|f| > B$. Taking a sample every $T_s$ seconds is the same as multiplying $g(t)$ by a train of unit impulses $\delta_{T_s}(t) = \sum_n \delta(t - nT_s)$, because an impulse picks out the value of $g$ at its own instant ($g(t)\delta(t - nT_s) = g(nT_s)\delta(t - nT_s)$):

$$\bar{g}(t) = g(t)\,\delta_{T_s}(t) = \sum_n g(nT_s)\,\delta(t - nT_s)$$

## Step 2 (Basic): Fourier series of the impulse train

The impulse train repeats every $T_s$, so it has a Fourier series with fundamental frequency $f_s = 1/T_s$:

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} D_n\, e^{jn2\pi f_s t}, \qquad D_n = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} \delta_{T_s}(t)\, e^{-jn2\pi f_s t}\, dt$$


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Inside the limits $-T_s/2$ to $T_s/2$ there is only one impulse, $\delta(t)$ at $t = 0$, and at $t = 0$ the exponential equals 1. An impulse picks out the value of the rest of the integrand at its position, so $D_n = \frac{1}{T_s} \cdot 1 = \frac{1}{T_s}$ for every $n$:

$$\delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn2\pi f_s t}$$

## Step 3 (Basic): Multiplying by an exponential shifts a spectrum

From the definition $G(f) = \int g(t)\, e^{-j2\pi ft}\, dt$:

$$\int_{-\infty}^{\infty} g(t)\, e^{j2\pi f_0 t}\, e^{-j2\pi ft}\, dt = \int_{-\infty}^{\infty} g(t)\, e^{-j2\pi (f-f_0)t}\, dt = G(f - f_0)$$

so $g(t)\, e^{j2\pi f_0 t} \Longleftrightarrow G(f - f_0)$: the spectrum moves to the right by $f_0$.

## Step 4: Spectrum of the sampled signal

Put the series of Step 2 into Step 1:

$$\bar{g}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} g(t)\, e^{jn2\pi f_s t}$$

Transform term by term, using Step 3 with $f_0 = nf_s$:

$$\bar{G}(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(f - nf_s)$$

The spectrum is $G(f)$ repeated at every multiple of $f_s$, each copy scaled by $1/T_s$.

## Step 5: When the copies do not overlap

The copy at $n = 0$ spans $-B$ to $B$. The next copy, centred at $f_s$, starts at $f_s - B$. They do not overlap if $f_s - B \geq B$, that is

$$f_s \geq 2B \quad \text{(Nyquist criterion)}$$

## Step 6: Pass through an LPF

Choose an ideal LPF with gain $T_s$ and cutoff $f_0$ such that $B \leq f_0 \leq f_s - B$ (above the top of the $n = 0$ copy, below the bottom of the $n = 1$ copy):

$$H(f) = T_s \,\text{rect}\left(\frac{f}{2f_0}\right)$$

It keeps only the $n = 0$ copy and cancels the $1/T_s$:

$$\bar{G}(f)H(f) = \frac{1}{T_s} G(f) \cdot T_s = G(f)$$

So the filter output is exactly $g(t)$. This proves the statement.


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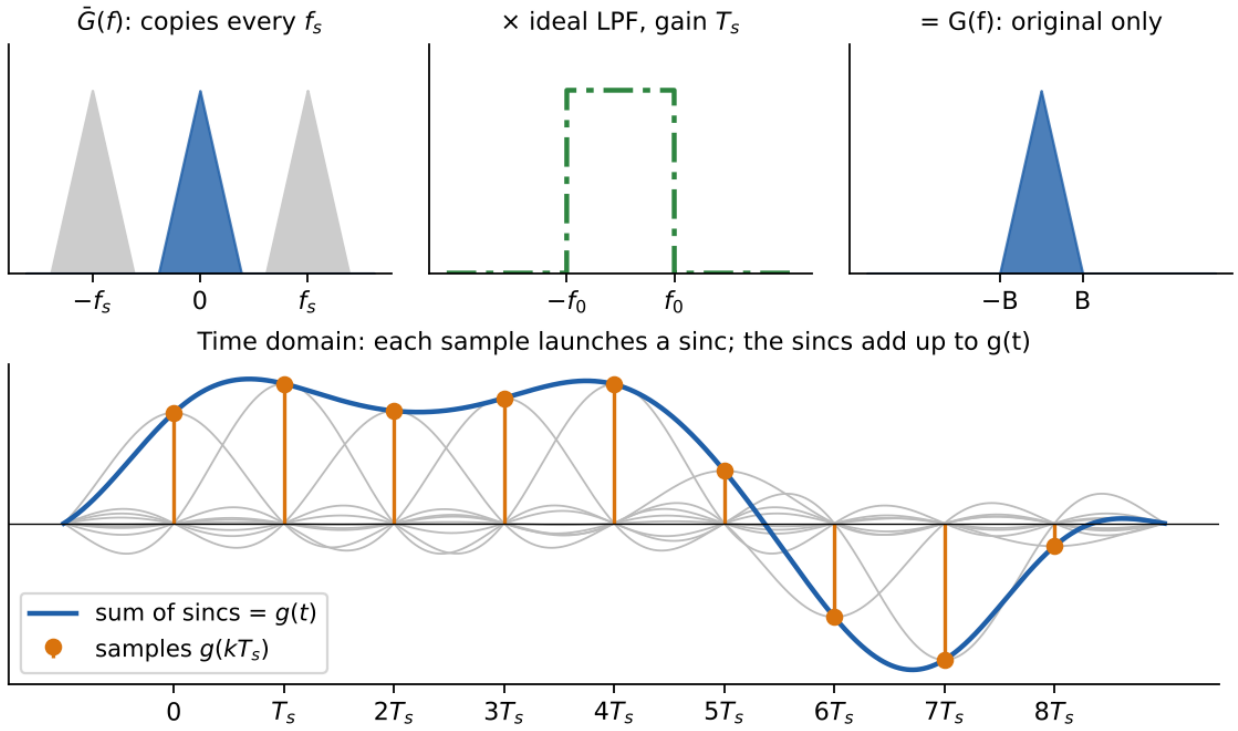



**Figure 15:** Reconstruction in frequency and time

**Reading the figure.** Top row, read as an equation: the sampled spectrum $\bar{G}(f)$ (blue copy at the centre, grey copies at $\pm f_s$) times the ideal LPF (green rectangle, cutoff $f_0$) equals $G(f)$, the blue copy alone. Bottom: in time, every sample (orange stem) launches a grey sinc pulse with its own height; each sinc is zero at every other sample instant, and the sum of all the sincs is the smooth signal $g(t)$ (thick blue), which passes through every sample.

## Step 7: The same in time (interpolation formula)

**Impulse response of the LPF.** Take Nyquist sampling, $f_s = 2B$, so $T_s = 1/2B$ and $f_0 = B$. The filter is $H(f) = T_s$ for $|f| \le B$ and 0 elsewhere. Its impulse response is the inverse Fourier transform:

$$h(t) = \int_{-B}^{B} T_s\, e^{j2\pi ft}\, df = T_s \left[\frac{e^{j2\pi ft}}{j2\pi t}\right]_{-B}^{B} = T_s\, \frac{e^{j2\pi Bt} - e^{-j2\pi Bt}}{j2\pi t} = T_s\, \frac{\sin(2\pi Bt)}{\pi t}$$

The last step uses $e^{jx} - e^{-jx} = 2j \sin x$. Multiply the top and bottom by $2B$ and write $\mathrm{sinc}(x) = \sin x / x$ (Lathi's convention):

$$h(t) = 2BT_s\, \frac{\sin(2\pi Bt)}{2\pi Bt} = 2BT_s\, \mathrm{sinc}(2\pi Bt) = \mathrm{sinc}(2\pi Bt)$$

since $2BT_s = 1$.

**Add the outputs.** The input to the filter is the impulse train of Step 1. Each impulse $g(kT_s)\delta(t - kT_s)$ produces the response $g(kT_s)\, h(t - kT_s)$, and the filter is linear, so the output is the sum:

$$g(t) = \sum_k g(kT_s)\, h(t - kT_s) = \sum_k g(kT_s)\, \mathrm{sinc}[2\pi B(t - kT_s)]$$

With $2\pi BkT_s = 2\pi Bk/(2B) = k\pi$:

$$\boxed{g(t) = \sum_k g(kT_s)\, \mathrm{sinc}(2\pi Bt - k\pi)}$$

Each sinc is 1 at its own sample instant and 0 at all other sample instants (where $2\pi Bt - k\pi$ is a non-zero multiple of $\pi$), so the sum passes through every sample and fills the gaps smoothly.

**Conditions for the LPF to work ("proper value"):**

- The signal is band-limited to $B$ and sampled at $f_s \geq 2B$.
- Cutoff $B \leq f_0 \leq f_s - B$, passband gain $T_s$.

**Practical Note:** an ideal LPF cannot be built (its sinc response starts before $t = 0$). So we sample a little faster than $2B$ (e.g. speech: $B = 3.4$ kHz, $f_s = 8$ kHz). The gap $f_s - 2B$ is a **guard band** where a real filter can roll off.

**4.(c) Write short note on: Companding, Information rate, and Entropy of a source. Marks: 5**

# 1. Theory

## 1. Why do we need companding? (Core Idea)

To send speech digitally, each sample is rounded to the nearest of a fixed set of levels. This is **quantization**. The rounding error is heard as noise. With equal steps of size $\Delta$, that noise is the same for every sample. A loud sound hides it well, but a soft sound does not. Most speech is soft, so we want small steps for small amplitudes.

## 2. What is information?

A message tells us more when it is less expected. "The sun rose today" tells us almost nothing, because we knew it would happen. So the information in a message of probability $p$ is defined as

$$I = \log_2 \frac{1}{p} \text{ bits}$$

A fair coin toss has $p = 1/2$, so it carries exactly 1 bit. A certain event has $p = 1$, so it carries none.

## 3. What are entropy and information rate?

A source sends many messages, each with its own probability. Its **entropy** $H$ is the average information per message: each $I$ weighted by how often it occurs. If the source sends $r$ messages each second, the **information rate** is simply $R = rH$ bits per second.

# 2. Answer

## i. Companding

**Problem:** a uniform quantizer has the same step size $\Delta$ everywhere, so the quantization noise $\Delta^2/12$ is the same for loud and soft signals. Speech is mostly soft, so soft talkers get a poor SQNR.

**Idea:** **compress** the signal before uniform quantization (boost small amplitudes, squeeze large ones), then **expand** it at the receiver with the inverse curve. Compressor + expander = **compander**. Effectively, small signals get small steps and large signals get large steps, so the SQNR becomes nearly the same for all signal levels.

- **μ-law** (North America, Japan, $\mu = 255$):

$$y = \frac{\ln(1 + \mu|m|/m_p)}{\ln(1 + \mu)} \operatorname{sgn}(m)$$

- **A-law** (Europe, Asia, $A = 87.6$): linear for $|m|/m_p \leq 1/A$, logarithmic above.
- With strong companding, SQNR $\approx \dfrac{3L^2}{[\ln(1 + \mu)]^2}$, independent of the signal power (about 38 dB for 8 bits).

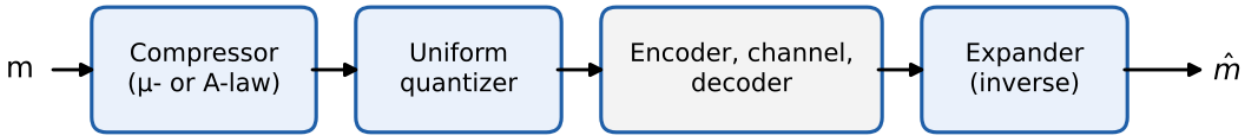



**Figure 16:** Compander chain

**Reading the figure.** From left to right: the compressor (μ-law or A-law) and a uniform quantizer together act as a non-uniform quantizer; then come the encoder, channel and decoder (grey); at the far end the expander, the exact inverse of the compressor, which restores the original amplitudes.

### ii. Information rate

If a source produces $r$ symbols per second and each symbol carries on average $H$ bits (its entropy), the **information rate** is

$$\boxed{R = r\,H \quad \text{bits/s}}$$

**Example 1**

a source sends 1000 symbols/s from an alphabet of four symbols with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$. $H = 1.75$ bits/symbol, so $R = 1000 \times 1.75 = 1750$ bits/s.

For error-free transmission the information rate must not exceed the channel capacity: $R \le C$.

### iii. Entropy of a source

The **entropy** is the average information per symbol of a source with symbol probabilities $P_i$:

$$H = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i} \quad \text{bits/symbol}$$

It is 0 when one symbol is certain and maximum, $\log_2 n$, when all $n$ symbols are equally likely. It is the minimum average number of bits per symbol needed to code the source (see 1(a) for the derivation).

**4.(d) Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them. Marks: 5**

# 1. Theory

## 1. Why convert digital data to an analog signal? (Core Idea)

Bits are a baseband signal. Radio channels and telephone lines pass only a band of frequencies around a carrier. So we let the bits change one property of a sinusoidal carrier $A\sin(2\pi ft + \phi)$. A sine has exactly three properties we can change: **amplitude** $A$, **frequency** $f$ and **phase** $\phi$. That gives the three basic mechanisms.

# 2. Answer

The three mechanisms are:

1. **Amplitude shift keying (ASK):** the bit changes the amplitude. Binary ASK/OOK: 1 → carrier on (or large), 0 → off (or small).
2. **Frequency shift keying (FSK):** the bit changes the frequency: 1 → $f_1$, 0 → $f_2$, with constant amplitude.
3. **Phase shift keying (PSK):** the bit changes the phase: BPSK 1 → 0°, 0 → 180°, i.e. $\pm A\sin\omega_c t$.

(A fourth, **QAM**, combines amplitude and phase changes.)

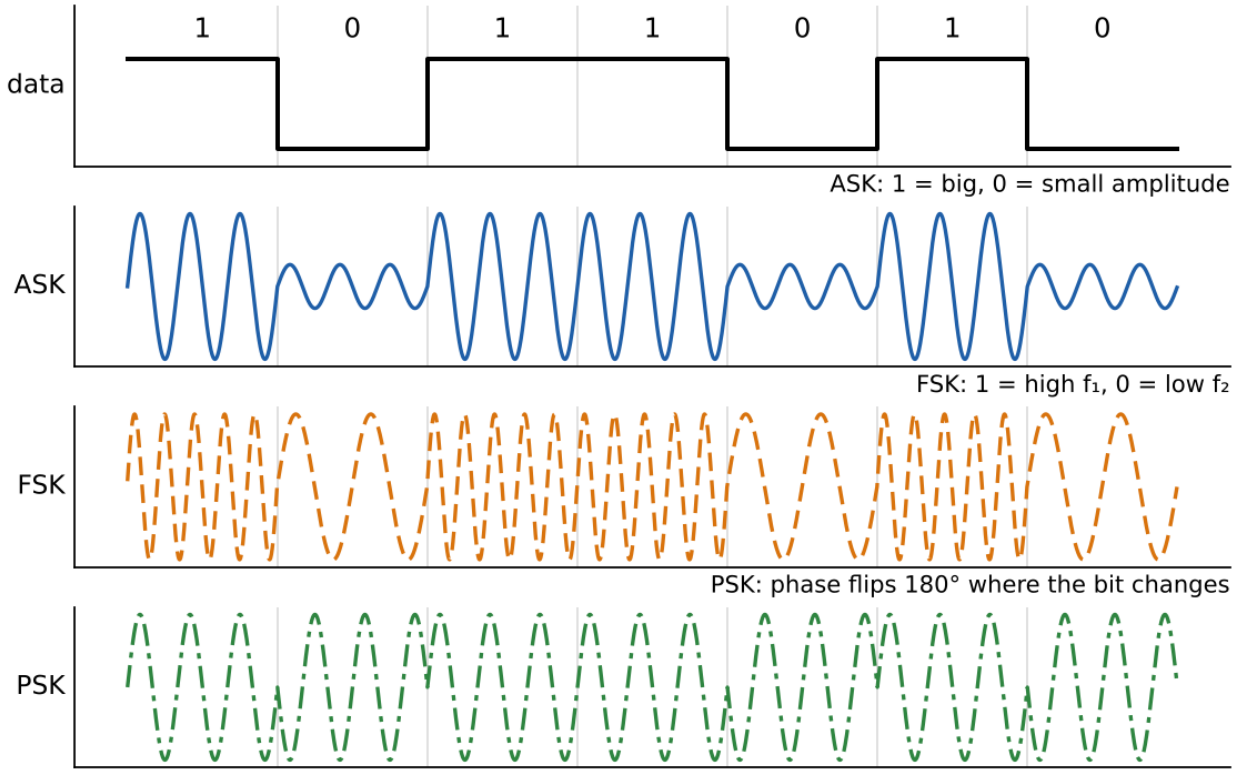



**Figure 17:** ASK, FSK and PSK waveforms for the same bits

**Reading the figure.** The top row is the data 1 0 1 1 0 1 0; grey lines mark the bit boundaries. In ASK the height of the wave changes (big for 1, small for 0). In FSK the height stays fixed and the oscillations are packed tightly for 1 ($f_1$) and spread out for 0 ($f_2$). In PSK amplitude and frequency stay the same and only the phase jumps by 180°, seen as a sudden turn of the wave wherever the bit changes.


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| Feature | ASK | FSK | PSK |
|---|---|---|---|
| Carrier property changed | amplitude | frequency | phase |
| Bandwidth (Forouzan) | $(1+d)S$ | $(1+d)S + 2\Delta f$ (largest) | $(1+d)S$ |
| Noise immunity | poor (noise affects amplitude) | better | best |
| Detection | coherent or envelope | coherent or non-coherent | coherent only (or DPSK) |
| Circuit complexity | simplest | moderate | most complex |
| Typical use | optical fibre (OOK), RFID | low-speed modems, Bluetooth (GFSK) | Wi-Fi, satellite, with QAM |

**Table 3:** Differences among ASK, FSK and PSK

where $S$ is the signal (baud) rate, $d$ the filter factor ($0 \le d \le 1$) and $\Delta f$ half the spacing between $f_1$ and $f_2$.

**4.(e) An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find- (i) Peak signal power to peak quantization noise power. (ii) System bandwidth. Marks: 5**

# 1. Theory

## 1. Step size and the quantization error (Core Idea)

A uniform quantizer divides the range $-m_p$ to $+m_p$ into $L$ equal steps of size $\Delta = 2m_p/L$. Every sample is rounded to the middle of its step, so the error is never larger than half a step: $|q| \le \Delta/2 = m_p/L$.

## 2. Bits and bandwidth

$L$ levels need $n = \log_2 L$ bits per sample. Sampling at $f_s$ gives a bit rate $R_b = nf_s$. A channel of bandwidth $B_T$ can carry at most $2B_T$ pulses per second (Nyquist), so the minimum bandwidth is $B_T = R_b/2$.

# 2. Answer

**Given:**

- Sampling rate $f_s = 8$ kHz $= 8000$ samples/s
- Number of levels $L = 32$

**Bits per sample:**

$$n = \log_2 L = \log_2 32 = 5 \text{ bits}$$

## Step 1: Peak signal power to peak quantization noise power

Peak signal amplitude $= m_p$; peak quantization error $= \Delta/2 = m_p/L$. Powers are proportional to amplitude squared:

$$\frac{\text{peak signal power}}{\text{peak noise power}} = \left(\frac{m_p}{m_p/L}\right)^2 = L^2 = 32^2 = 1024$$

$$\boxed{\frac{S_{peak}}{N_{peak}} = 1024 = 30.1 \text{ dB}}$$

## Step 2: System bandwidth

$$R_b = nf_s = 5 \times 8000 = 40000 \text{ bits/s} = 40 \text{ kbit/s}$$


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$$B_T = \frac{R_b}{2} = \frac{40000}{2} = 20000 \text{ Hz}$$

$$\boxed{B_T = 20 \text{ kHz (minimum)}}$$

**Final Answer:** (i) $S_{peak}/N_{peak} = L^2 = 1024$ (30.1 dB); (ii) $R_b = 40$ kbit/s and minimum bandwidth $B_T = 20$ kHz.

# 3. Notes and Remarks

- Many textbooks take "noise power" as the **average** quantization noise power $\Delta^2/12 = m_p^2/(3L^2)$. Then the peak-signal-to-noise ratio is $3L^2 = 3072$ (34.9 dB). Say which one you use in the exam.
- Lathi gives the minimum PCM transmission bandwidth as $B_T = nB$, which is $R_b/2$ at Nyquist sampling. With $f_s = 8$ kHz the audio band is at most 4 kHz, and $nB = 5 \times 4 = 20$ kHz agrees. A practical system with non-ideal pulses needs more (up to $R_b = 40$ kHz).

# 4. Final Concise Answer

**(a)** PCM = sampling ($f_s \geq 2B$) + quantizing ($L$ levels) + encoding ($n = \log_2 L$ bits); transmitted as pulses with regenerative repeaters; receiver: regenerate, decode, LPF.

**(b)** $\bar{G}(f) = \frac{1}{T_s}\sum G(f - nf_s)$; if $f_s \geq 2B$, an LPF with gain $T_s$ and cutoff $B \leq f_0 \leq f_s - B$ keeps only $G(f)$, giving $g(t) = \sum g(kT_s)\text{sinc}(2\pi Bt - k\pi)$.

**(c)** Companding: compress, quantize uniformly, expand (μ-law $\mu = 255$, A-law $A = 87.6$) for nearly constant SQNR. Information rate $R = rH$ bits/s. Entropy $H = \sum P_i \log_2(1/P_i)$ bits/symbol.

**(d)** ASK (amplitude), FSK (frequency), PSK (phase); FSK needs the most bandwidth, PSK has the best noise immunity, ASK is the simplest.

**(e)** $n = 5$; $S_{peak}/N_{peak} = L^2 = 1024$ (30.1 dB) [$3L^2 = 3072$, 34.9 dB, with average noise]; $R_b = 40$ kbit/s, $B_T = 20$ kHz.


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5. (a) Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM. Marks: 6
   (b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. Marks: 7
   (c) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'. Marks: 6
   (d) Briefly describe coherent and non-coherent detection of BASK and BFSK. Marks: 6

**5.(a) Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM. Marks: 6**

# 1. Theory

## 1. Delta modulation in one paragraph (Core Idea)

**Delta modulation (DM)** sends only one bit per sample: 1 if the signal is above the previous approximation (step up by $\Delta$), 0 if below (step down). The receiver adds up these steps with an **integrator** to rebuild a staircase that follows the signal. So DM sends the *changes* (the slope) of the signal, not its value.

## 2. The two problems of DM

- **Errors accumulate:** the receiver's integrator adds up every bit it gets, including wrong ones. One wrong bit shifts the staircase for ever after.
- **Slope overload:** if the signal rises faster than $\Delta$ per sample, the staircase cannot keep up. DM also handles dc and slow signals badly.

**Sigma-delta ($\Sigma$-$\Delta$)** fixes both by moving the integrator from the receiver into the transmitter. "$\Sigma$" (sum) means integration; "$\Delta$" means the delta modulator.

# 2. Answer

**Principle:** Because the system is linear, the integrator at the receiver of a DM system can be moved to the input of the transmitter: integrate the message first, then delta-modulate it. The transmitter's own feedback integrator and this new input integrator can then be merged into a single integrator placed after the subtractor. The result is the $\Sigma$-$\Delta$ modulator:

$$m(t) \to \Sigma \text{ (subtract output)} \to \int \to \text{comparator } (\pm E) \to \text{sampler/pulse generator} \to \text{output}$$

with the output fed back **directly** (no integrator) to the subtractor. The receiver is **only a low-pass filter**.


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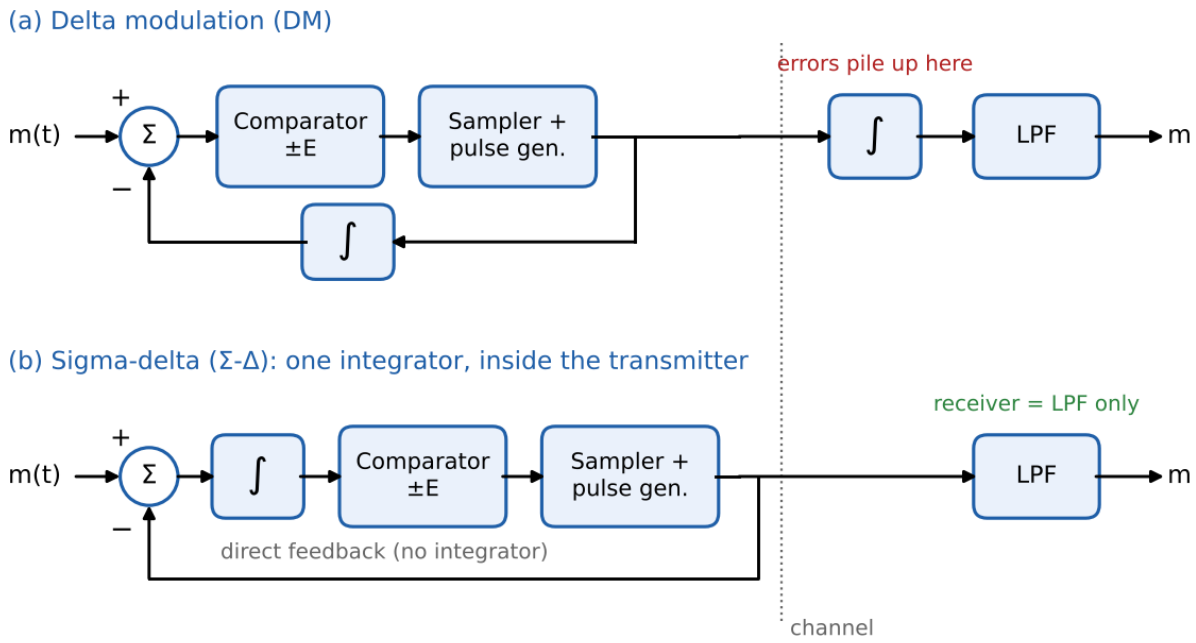



**Figure 18:** Delta modulation and sigma-delta modulation compared

**Reading the figure.** The dotted line is the channel: the transmitter is on its left and the receiver on its right. (a) Ordinary DM has an integrator in the feedback path and another one at the receiver, where channel errors pile up. (b) In Σ-Δ the only integrator sits right after the subtractor inside the transmitter, the feedback is a plain wire, and the receiver is just an LPF.

**How it works:** the integrator accumulates the error $\int (m - \hat{m})\, dt$ between the message and the output pulses. The comparator outputs $+E$ when this accumulated error is positive and $-E$ when it is negative, which pushes the error back towards zero. So the *density* of $+E$ pulses follows the message *amplitude*: a high signal gives mostly $+E$ pulses, a low signal mostly $-E$ pulses. The receiver's LPF simply averages the pulses and recovers $m(t)$.

**Advantages over DM:**

1. **Simple receiver:** only an LPF; no integrator.
2. **No accumulation of channel errors:** with no integrator at the receiver, a wrong bit affects only a tiny part of the average, and the effect does not last.
3. **No slope overload:** the bits represent the amplitude, not the slope, so fast-changing signals are not a problem (the limit becomes the amplitude, $|m| \leq E$).
4. **dc and low frequencies pass correctly**, which DM cannot do well.
5. **Noise shaping:** with heavy oversampling, the loop pushes the quantization noise to high frequencies, where the LPF removes it. This gives high resolution from a 1-bit quantizer, which is why almost all audio ADCs and DACs are Σ-Δ.

**5.(b) With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks. Marks: 7**

# 1. Theory

## 1. Two signals on one carrier (Core Idea)

$\cos \omega_c t$ and $\sin \omega_c t$ have the same frequency but are 90° apart. They are **orthogonal**: the average of their product is zero. So we can put one message on the cosine and another on the sine, send them together in the *same* band, and still separate them at the receiver. This is **quadrature amplitude modulation (QAM)**. It doubles the use of the bandwidth.


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## 2. Why "coherent"?

To pull one message out, the receiver multiplies by the matching carrier. The carrier must have exactly the right frequency and phase; otherwise the cosine branch also picks up part of the sine message.

# 2. Answer

**QAM signal:**

$$\varphi_{QAM}(t) = m_1(t)\cos\omega_c t + m_2(t)\sin\omega_c t$$

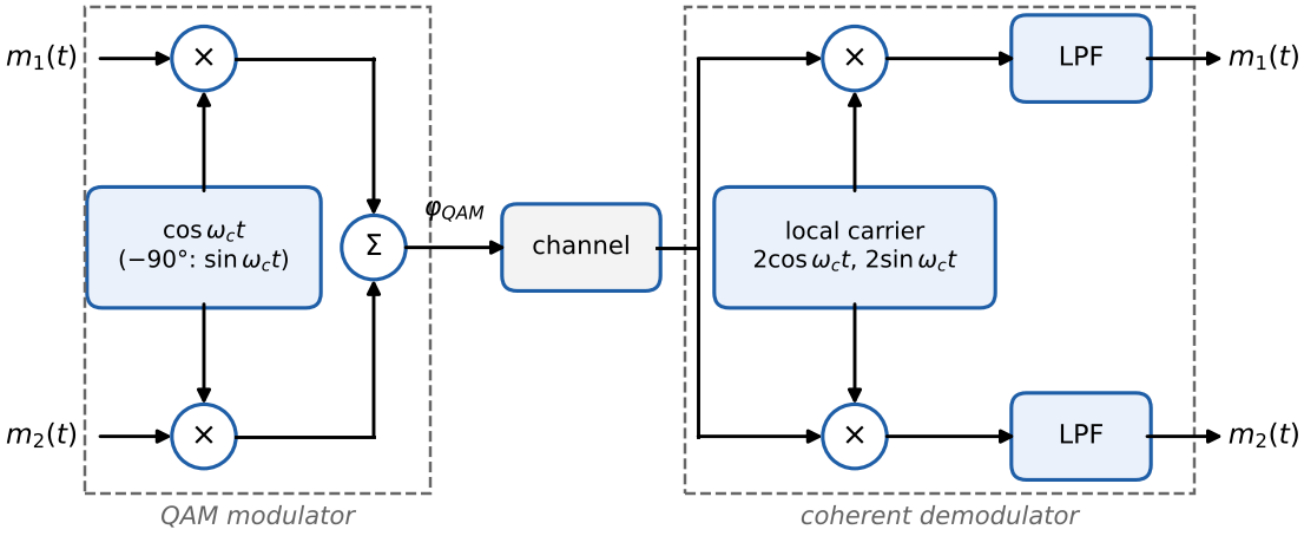



**Figure 19:** QAM modulator and coherent demodulator

**Reading the figure.** Left: $m_1$ multiplies $\cos\omega_c t$ and $m_2$ multiplies $\sin\omega_c t$ (the carrier shifted by 90°); the two products are added and sent. Right: the receiver splits the signal into two branches, multiplies one by $2\cos\omega_c t$ and the other by $2\sin\omega_c t$, and low-pass filters each, giving $m_1(t)$ and $m_2(t)$.
The three identities used below are

$$2\cos^2 x = 1 + \cos 2x, \qquad 2\sin^2 x = 1 - \cos 2x, \qquad 2\sin x\cos x = \sin 2x$$

**Upper (in-phase) branch:** multiply the received signal by $2\cos\omega_c t$ and multiply out:

$$x_1(t) = 2\varphi_{QAM}(t)\cos\omega_c t = m_1(t)\cdot 2\cos^2\omega_c t + m_2(t)\cdot 2\sin\omega_c t\cos\omega_c t$$

Apply the first and third identities with $x = \omega_c t$:

$$x_1(t) = m_1(t)[1+\cos 2\omega_c t] + m_2(t)\sin 2\omega_c t = m_1(t) + m_1(t)\cos 2\omega_c t + m_2(t)\sin 2\omega_c t \xrightarrow{\text{LPF}} m_1(t)$$

**Lower (quadrature) branch:** multiply by $2\sin\omega_c t$:

$$x_2(t) = 2\varphi_{QAM}(t)\sin\omega_c t = m_1(t)\cdot 2\sin\omega_c t\cos\omega_c t + m_2(t)\cdot 2\sin^2\omega_c t$$

Apply the third and second identities:

$$x_2(t) = m_1(t)\sin 2\omega_c t + m_2(t)[1-\cos 2\omega_c t] = m_2(t) - m_2(t)\cos 2\omega_c t + m_1(t)\sin 2\omega_c t \xrightarrow{\text{LPF}} m_2(t)$$

The terms at $2\omega_c$ are removed by the LPFs, and because of orthogonality each branch keeps only its own message.

**Effect of a phase error:** let the upper local carrier be $2\cos(\omega_c t+\theta)$. Then

$$2\varphi_{QAM}(t)\cos(\omega_c t+\theta) = m_1(t)\cdot 2\cos\omega_c t\cos(\omega_c t+\theta) + m_2(t)\cdot 2\sin\omega_c t\cos(\omega_c t+\theta)$$


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Use $2\cos A\cos B = \cos(A-B)+\cos(A+B)$ and $2\sin A\cos B = \sin(A-B)+\sin(A+B)$, with $A=\omega_c t$, $B=\omega_c t+\theta$, so $A-B=-\theta$ and $A+B=2\omega_c t+\theta$. Since $\cos(-\theta)=\cos\theta$ and $\sin(-\theta)=-\sin\theta$:

$$= m_1(t)[\cos\theta + \cos(2\omega_c t+\theta)] + m_2(t)[-\sin\theta + \sin(2\omega_c t+\theta)]$$

The LPF removes the terms at $2\omega_c$, so the upper output becomes

$$m_1(t)\cos\theta - m_2(t)\sin\theta$$

so part of $m_2$ leaks into the $m_1$ output (**co-channel interference** or crosstalk). A frequency error makes it worse. Hence the local carriers must be exactly synchronized.

**Advantages:**

- Two messages in the bandwidth of one DSB signal: **double the bandwidth efficiency** (same as SSB, but without the sharp filters).
- In digital form (M-QAM: 16-, 64-, 256-QAM) many bits per symbol, so high data rates in a narrow band (Wi-Fi, cable modems, 4G/5G, digital TV).
- Good noise performance for a given bandwidth compared with M-ASK.

**Drawbacks:**

- Needs exact carrier **phase and frequency synchronization** (pilot or carrier-recovery loop), so the receiver is complex.
- Any phase or frequency error causes crosstalk between the two channels.
- Amplitude carries information, so amplifiers must be linear (sensitive to non-linearity and fading); dense constellations need a high SNR.

**5.(c) Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'. Marks: 6**

# 1. Theory

## 1. The idea of DPSK (Core Idea)

Ordinary PSK puts each bit in the absolute phase of the carrier, so the receiver needs a perfect local carrier. **Differential PSK (DPSK)** puts each bit in the *change* of phase: data 1 → flip the phase, data 0 → keep the phase. The receiver then only compares each bit's signal with the previous one. No local carrier is needed.

## 2. The stages

- **Differential encoder:** $q_k = I_k \oplus q_{k-1}$ (XOR with the previous encoded bit). Since $x\oplus 1=\bar{x}$ and $x\oplus 0=x$: 1 → flip, 0 → keep.
- **Polar coder:** $q_k=1 \to a_k=+1$ (phase 0), $q_k=0 \to a_k=-1$ (phase $\pi$).
- **PSK modulator:** $\varphi(t) = a_k A_c\cos\omega_c t$.
- **Receiver:** multiply the signal by itself delayed by one bit ($T_b$), then LPF: output $\propto a_k a_{k-1}$. Positive (same phase) → 0; negative (phase changed) → 1.

# 2. Answer

**Given:** message $I_k$ = 1 0 1 0 0 1 0 1 1 0 (10 bits).

**Convention (Lathi's differential encoding, a 1 flips the phase):** $q_k = I_k \oplus q_{k-1}$, with a reference bit $q_0=0$ sent first (phase $\pi$), so that the first data bit also has a reference.

| $k$ | 0 (ref) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Data $I_k$ | – | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| $q_{k-1}$ (delayed) | – | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| Encoded $q_k = I_k \oplus q_{k-1}$ | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |


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| $k$ | 0 (ref) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Polar $a_k$ | $-1$ | $+1$ | $+1$ | $-1$ | $-1$ | $-1$ | $+1$ | $+1$ | $-1$ | $+1$ | $+1$ |
| Phase $\theta_k$ | $\pi$ | 0 | 0 | $\pi$ | $\pi$ | $\pi$ | 0 | 0 | $\pi$ | 0 | 0 |
| Rx: delayed $a_{k-1}$ | – | $-1$ | $+1$ | $+1$ | $-1$ | $-1$ | $-1$ | $+1$ | $+1$ | $-1$ | $+1$ |
| Rx: LPF output sign $a_k a_{k-1}$ | – | $-$ | $+$ | $-$ | $+$ | $+$ | $-$ | $+$ | $-$ | $-$ | $+$ |
| Phase change? | – | yes | no | yes | no | no | yes | no | yes | yes | no |
| Detected bit | – | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |

**Table 4:** DPSK encoder and receiver for the message 1010010110

**Check:** the detected bits 1 0 1 0 0 1 0 1 1 0 equal the message. ✔
**How to fill it in the exam:** work column by column from left to right. Take the previous $q$, XOR it with the data bit to get the new $q$, convert to $\pm 1$ and phase; at the receiver, multiply the current $a_k$ by the previous one: "$-$" means the phase changed (bit 1), "$+$" means it did not (bit 0).

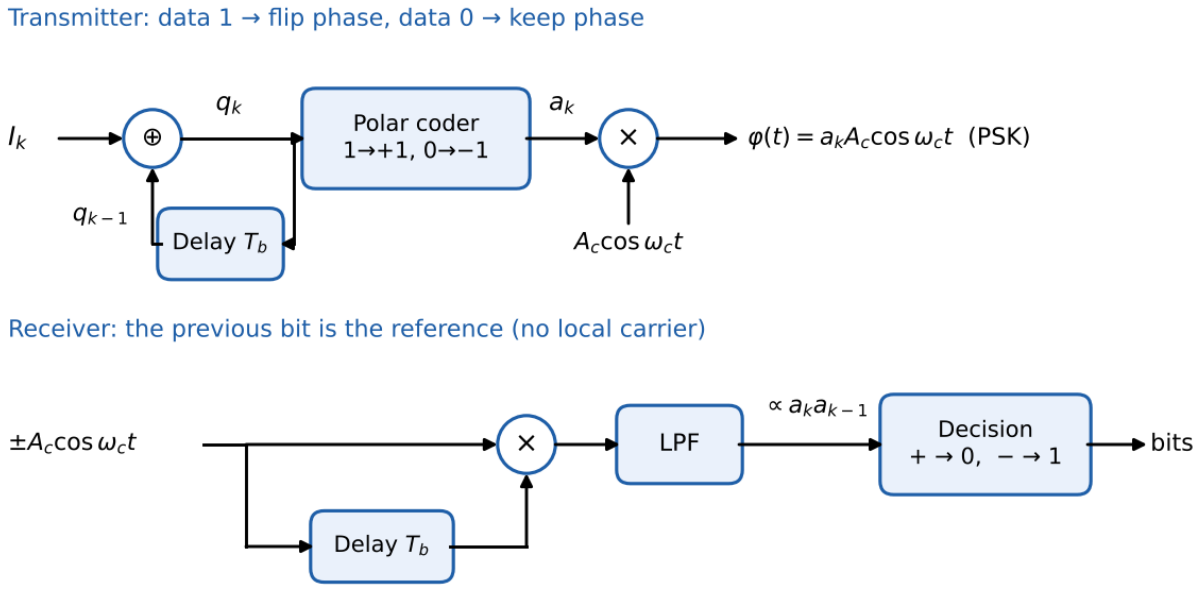



**Figure 20:** DPSK transmitter and receiver

**Reading the figure.** Top: the differential encoder (XOR $\oplus$ of $I_k$ with $q_{k-1}$, which comes back through a one-bit delay), the polar coder and the multiplier with $A_c \cos \omega_c t$. Bottom: the receiver multiplies the incoming signal by a one-bit-delayed copy of itself, low-pass filters the product (output $\propto a_k a_{k-1}$) and decides: positive $\to$ 0, negative $\to$ 1.

# 3. Notes and Remarks

- Some books start with reference bit $q_0 = 1$, or use XNOR ($q_k = \overline{I_k \oplus q_{k-1}}$, so a 0 flips the phase). The $q_k$ and phase rows then change, but the detected bits are the same. State your convention.
- A single bit error in DPSK usually causes *two* detected errors, because each received bit is also the reference for the next one.

**5.(d) Briefly describe coherent and non-coherent detection of BASK and BFSK. Marks: 6**

# 1. Theory

## 1. Two kinds of receiver (Core Idea)

- A **coherent** receiver makes its own copy of the carrier (same frequency *and* phase), multiplies, and low-pass filters. It performs best in noise but needs carrier recovery.

- A **non-coherent** receiver uses only the **envelope** (or the energy in a band), so it needs no local carrier. It is simpler and cheaper but slightly worse in noise.

BASK has information in its amplitude and BFSK in which frequency is present, so both can be detected either way. (BPSK cannot be detected non-coherently: its envelope is constant.)

## 2. Answer

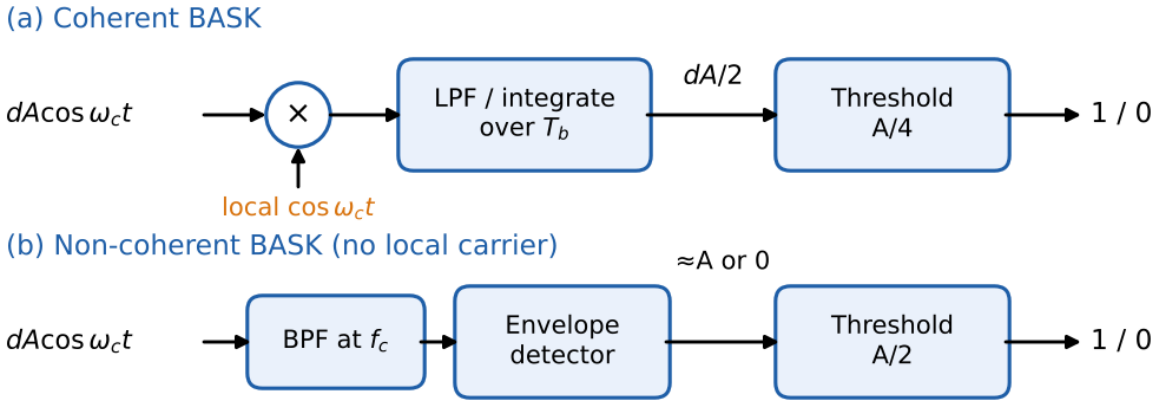



**Figure 21:** Coherent and non-coherent detection of BASK

**Reading the figure.** (a) Coherent BASK: multiply by a local carrier, low-pass filter (or integrate over one bit) and compare $dA/2$ with the threshold $A/4$. (b) Non-coherent BASK: band-pass filter, envelope detector (output about A or 0) and a threshold at $A/2$; no local carrier.

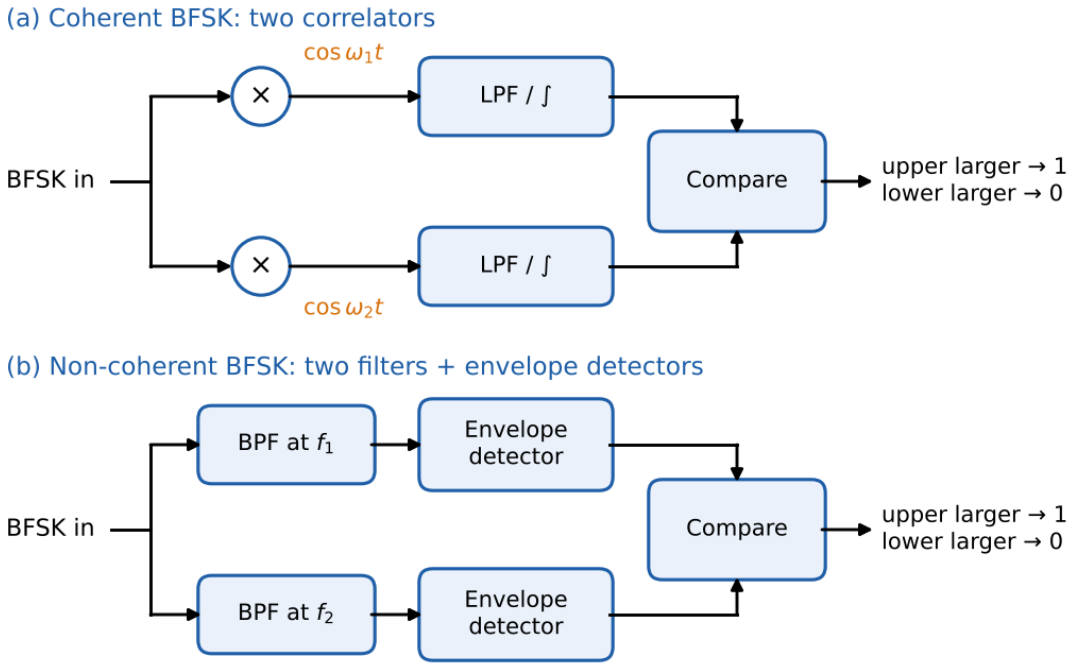



**Figure 22:** Coherent and non-coherent detection of BFSK

**Reading the figure.** Both BFSK receivers have two branches, one for each frequency, and a comparator that picks the larger branch. (a) Coherent: each branch multiplies by its own local carrier ($\cos\omega_1 t$ or $\cos\omega_2 t$) and integrates. (b) Non-coherent: each branch is a band-pass filter at $f_1$ or $f_2$ followed by an envelope detector.


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### i. Coherent detection of BASK

Received $s(t) = d\,A\cos\omega_c t$, with $d = 1$ or 0. Multiply by the local carrier $\cos\omega_c t$:

$$dA\cos^2\omega_c t = \frac{dA}{2}(1+\cos 2\omega_c t) \xrightarrow{\text{LPF / integrate over } T_b} \frac{dA}{2}$$

The decision device compares $dA/2$ with a threshold $A/4$: above $\to$ 1, below $\to$ 0. A phase error $\theta$ reduces the output by $\cos\theta$, so carrier synchronization is required.

### ii. Non-coherent detection of BASK

The received signal goes through a BPF (around $f_c$) and an **envelope detector**. The envelope is about $A$ during a 1 and about 0 during a 0. A comparator with a threshold halfway gives the bits. No local carrier is needed.

### iii. Coherent detection of BFSK

Two correlator branches: one multiplies by $\cos\omega_1 t$, the other by $\cos\omega_2 t$, each followed by an LPF (integrator over $T_b$). When $f_1$ is received, the upper branch gives about $A/2$ and the lower about 0 (the frequencies differ, so the product averages to zero). A comparator chooses the larger output: upper $\to$ 1, lower $\to$ 0. Both local carriers must be phase-locked.

### iv. Non-coherent detection of BFSK

Two **band-pass filters**, one tuned to $f_1$ and one to $f_2$, each followed by an envelope detector. Only the filter matching the received tone gives a large envelope. A comparator picks the larger: 1 if the $f_1$ branch wins, 0 if the $f_2$ branch wins.

| | Coherent | Non-coherent |
|---|---|---|
| Local carrier | needed (frequency and phase) | not needed |
| Main blocks | multiplier + LPF/integrator + decision | BPF + envelope detector + decision |
| Noise performance | better (lower bit error rate) | slightly worse |
| Complexity | high (carrier recovery) | low |

**Table 5:** Coherent versus non-coherent detection

## 3. Final Concise Answer

**(a)** $\Sigma$-$\Delta$ = DM with the receiver's integrator moved to the transmitter input and merged into one integrator after the subtractor; comparator output density follows the signal amplitude; receiver is just an LPF. Better than DM: no error accumulation, no slope overload, dc passes, simple receiver, noise shaping with oversampling.

**(b)** $\varphi_{QAM} = m_1\cos\omega_c t + m_2\sin\omega_c t$; multiply by $2\cos\omega_c t$ and $2\sin\omega_c t$, LPF, get $m_1$ and $m_2$. Advantage: double bandwidth efficiency, high data rates. Drawback: needs exact carrier synchronization; a phase error gives $m_1\cos\theta - m_2\sin\theta$ (crosstalk); needs linear amplifiers.

**(c)** With $q_0 = 0$: $q_k = 0\ 1\ 1\ 0\ 0\ 0\ 1\ 1\ 0\ 1\ 1$, $a_k = -1\ +1\ +1\ -1\ -1\ -1\ +1\ +1\ -1\ +1\ +1$; receiver signs $-\ +\ -\ +\ +\ -\ +\ -\ -\ +$; detected 1010010110 ✔.

**(d)** BASK: coherent (multiply by carrier, LPF, threshold) or envelope detector + threshold. BFSK: coherent (two correlators at $f_1$, $f_2$, pick larger) or two BPFs + envelope detectors, pick larger. Coherent is better in noise but needs carrier recovery.


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6. (a) Write short notes on: Multilevel multiplexing and Multiple slot allocation. Marks: 6

(b) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. Marks: 6

(c) Compare between FDMA and CDMA. Marks: 3

(d) Discuss the design procedures of a communication system. Marks: 7

(e) Differentiate between multiplexing and multiple access technologies. Marks: 3

**6.(a) Write short notes on: Multilevel multiplexing and Multiple slot allocation. Marks: 6**

# 1. Theory

## The basics

**Multiplexing** sends several signals over one link. In **time-division multiplexing (TDM)**, the link is divided into repeating time slots, and each input owns a slot in every round.

## 1. The problem: inputs with different rates (Core Idea)

Synchronous TDM takes one unit from each input in turn, round and round. That works when every input has the same data rate. If one input is faster, it produces data faster than its single slot can carry, and it falls behind. Two simple fixes work when the rates are related by whole numbers.

# 2. Answer

## i. Multilevel multiplexing

Used when one input's rate is a **multiple** of the others. The slower inputs are first multiplexed together into one faster stream, which is then multiplexed with the faster inputs, in two (or more) levels.

**Example 1**

inputs of 20, 20 and 40 kbps. The two 20 kbps inputs are combined by a first multiplexer into one 40 kbps stream; a second multiplexer combines it with the 40 kbps input, giving $2 \times 40 = 80$ kbps on the link.

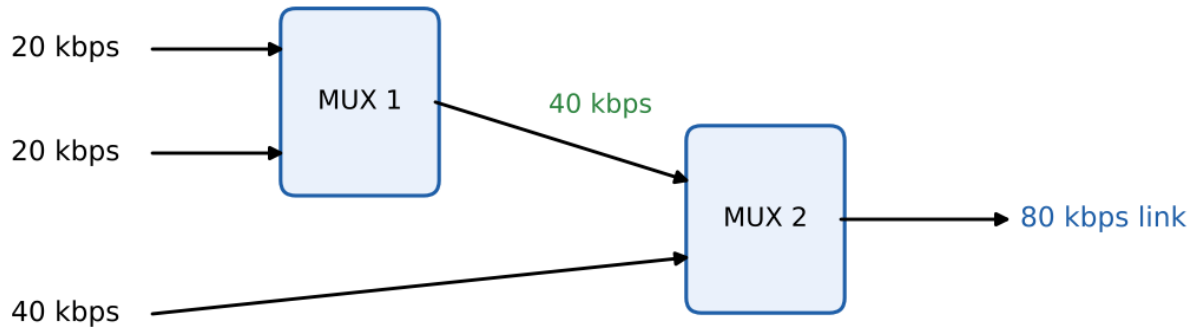



**Figure 23:** Multilevel multiplexing

**Reading the figure.** MUX 1 combines the two 20 kbps inputs into a 40 kbps stream (labelled in green); MUX 2 combines this with the other 40 kbps input, and the link carries 80 kbps.

### ii. Multiple-slot allocation

Used when the rates share a common unit. Instead of one slot, a faster input is given **several slots per frame**, in proportion to its rate. Inside the multiplexer a serial-to-parallel converter splits the fast input into several lines, each with the base rate.

**Example 2**

inputs of 5, 10 and 15 kbps. The base unit is 5 kbps, so they get 1, 2 and 3 slots. A frame has 6 slots, and the link rate is $6 \times 5 = 30$ kbps.

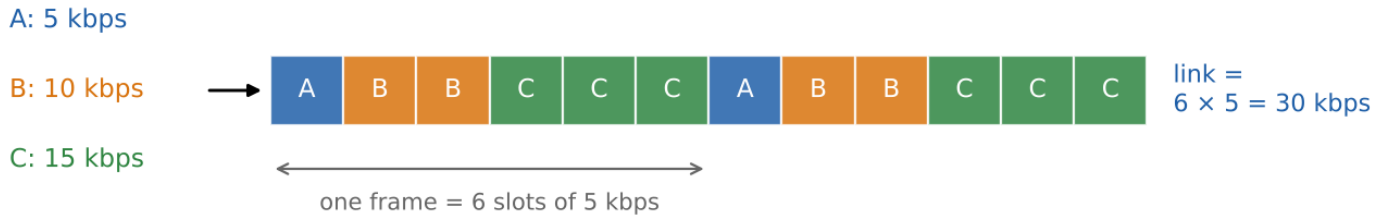



**Figure 24:** Multiple-slot TDM

**Reading the figure.** Input A (5 kbps, blue) gets one slot, B (10 kbps, orange) two and C (15 kbps, green) three in each frame of six slots, each slot labelled with its input's letter; two frames are shown. Each slot carries 5 kbps, so the link carries $6 \times 5 = 30$ kbps.

**Key Points:** in both methods the link rate equals the sum of the input rates (no waste). When rates are not whole multiples, a third method, **pulse stuffing** (adding dummy bits to the slower input), is used.

**6.(b) It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data. Marks: 6**

# 1. Theory

## 1. The plan (Core Idea)

We follow the bits: (1) how many samples per second per patient, (2) how many bits per sample (from the error limit), (3) bits per second for all patients, (4) the minimum bandwidth that can carry those bits.

## 2. The error limit decides the number of levels

A uniform quantizer with $L$ levels over $-m_p$ to $+m_p$ has step $\Delta = 2m_p/L$. The largest rounding error is half a step: $\Delta/2 = m_p/L$. Making this small enough fixes $L$, and $L = 2^n$ fixes the number of bits $n$.

## 3. Minimum bandwidth

A channel of bandwidth $B_T$ can carry at most $2B_T$ pulses per second (Nyquist), so $B_T = R_b/2$.

# 2. Answer

**Given:**


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- Number of patients (channels) = 10
- ECG bandwidth $B = 100$ Hz
- Maximum sample error = 0.25% = 0.0025 of the peak amplitude $m_p$
- Sampling rate $\geq 2\times$ Nyquist rate

### Step 1: Sampling rate

$$\text{Nyquist rate} = 2B = 2 \times 100 = 200 \text{ Hz}$$

$$f_s = 2 \times 200 = 400 \text{ samples/s per patient}$$

### Step 2: Number of levels and bits

$$\frac{\Delta}{2} = \frac{m_p}{L} \leq 0.0025\, m_p \quad \Rightarrow \quad L \geq \frac{1}{0.0025} = 400$$

$L$ must be a power of 2: $2^8 = 256$ is too small, so

$$L = 2^9 = 512, \qquad n = 9 \text{ bits/sample}$$

(Check: maximum error = $m_p/512 = 0.195\%$ of $m_p$, within 0.25%.)

### Step 3: Bit rate of the multiplexed data

$$R_b = (\text{patients}) \times f_s \times n = 10 \times 400 \times 9 = 36000 \text{ bits/s} = 36 \text{ kbit/s}$$

### Step 4: Minimum cable bandwidth

$$B_T = \frac{R_b}{2} = \frac{36000}{2} = 18000 \text{ Hz}$$

$$\boxed{B_T = 18 \text{ kHz}}$$

**Final Answer:** 9 bits/sample, 400 samples/s per patient, 36 kbit/s in total; minimum cable bandwidth **18 kHz**.

## 3. Notes and Remarks

- Framing and synchronization bits are ignored, as the question asks for the minimum.
- A common slip is to use $\Delta \leq 0.0025 m_p$ instead of $\Delta/2$; that would give $L \geq 800$, $n = 10$ and 20 kHz. The *error* of a rounding quantizer is at most half a step.

**6.(c) Compare between FDMA and CDMA. Marks: 3**

## 1. Theory

### 1. What is multiple access? (Core Idea)

Many users must share one radio channel at once. If they all sent the same signal in the same band at the same time, a receiver could not tell them apart. So each user needs something that separates it from the others. There are three things to divide: frequency, time and code.

### 2. How does FDMA separate users?

Frequency division multiple access gives each user its own narrow band, like giving each conversation its own room. A receiver tunes to one band with a filter. Small empty **guard bands** stop neighbours overlapping. The number of bands is fixed, so when they are all taken the next user is refused.

### 3. How does CDMA separate users?

Code division multiple access lets everyone use the whole band at the same time, like people talking in one room in different languages. Each user multiplies its bits by a fast, unique **code**, which spreads the signal over a wide band. The receiver multiplies by the same code again. The wanted signal is recovered, while the others, having different codes, average to nearly zero and look like weak noise. Each extra user adds a little more noise, so capacity has a soft limit.

## 2. Answer

| Feature | FDMA | CDMA |
|---|---|---|
| Users separated by | frequency: each user has its own band | code: each user has its own spreading code |
| Band and time | narrow band, all the time | the whole wide band, all the time |
| Based on | FDM | spread spectrum (DSSS) |
| Guard | guard bands between channels | no guard bands; codes must be orthogonal (or nearly) |
| Capacity limit | fixed by number of bands (hard limit) | limited by interference (soft limit; more users raise the noise) |
| Interference / security | prone to adjacent-channel interference; easy to intercept | resistant to jamming and multipath; hard to intercept |
| Power control | not critical | essential (near-far problem) |
| Example | 1G AMPS | IS-95, 3G (WCDMA) |

**Table 6:** Comparison of FDMA and CDMA

**6.(d) Discuss the design procedures of a communication system. Marks: 7**

## 1. Theory

### 1. What does a communication system contain? (Core Idea)

Every system has a **source**, a **transmitter** (which prepares the message for the channel), a **channel** (which adds noise, distortion and attenuation), a **receiver** (which undoes the transmitter's work) and a **destination**. Designing it means choosing each block so that the message arrives with the required quality, using the bandwidth and power we are allowed, at a reasonable cost.

```
Source -> Source encoder -> Channel encoder -> Modulator
                                                  |
                                                  v
                          noise, interference -> CHANNEL
                                                  |
                                                  v
Destination <- Source decoder <- Channel decoder <- Demodulator
```

**Listing 1:** Block diagram of a digital communication system

## 2. Answer

The design is done step by step:

1. **Define the requirements:** the type of message (voice, video, data), the data rate or message bandwidth, the required quality (SNR for analog, bit error rate for digital), the distance, the number of users, mobility, delay limits and cost.
2. **Study the channel:** wire, optical fibre or radio; available bandwidth $B$; attenuation and noise level; interference; fading and multipath. Regulations decide the allowed frequency band and power.
3. **Check the theoretical limit:** with Shannon's formula $C = B\log_2(1 + S/N)$, check that the required rate is possible ($R \leq C$) and find the minimum SNR needed.
4. **Source coding:** digitize (sampling, quantization, PCM/DPCM/$\Sigma$-$\Delta$) and compress to remove redundancy, so as few bits as possible are sent.
5. **Channel coding:** add controlled redundancy (error-detecting and correcting codes) to meet the error-rate target at the available SNR.


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6. **Choose the modulation:** trade bandwidth against power. Bandwidth-limited channels use M-ary schemes (QPSK, QAM); power-limited channels use robust schemes (BPSK, FSK, spread spectrum). Choose coherent or non-coherent detection.
7. **Multiplexing and multiple access:** FDM/TDM/WDM to share a link; FDMA/TDMA/CDMA for many users of one medium.
8. **Link (power) budget:** transmitter power + antenna gains − path loss − other losses = received power; compare it with the receiver noise to get the SNR, and keep a margin for fading.
9. **Design the hardware:** antennas, amplifiers, filters, oscillators, synchronization (carrier and timing recovery), equalizers.
10. **Simulate, build and test:** check the bandwidth, SNR or bit error rate and the cost against the requirements, then refine.

**Key trade-off:** the three resources are **bandwidth**, **power** and **complexity (cost)**. Improving the quality with one usually costs another, e.g. PCM trades bandwidth for SNR, M-ary QAM trades power for bandwidth.

**6.(e) Differentiate between multiplexing and multiple access technologies. Marks: 3**

# 1. Theory

## 1. Why share a channel at all? (Core Idea)

A cable or a radio band is expensive, and it often has more capacity than one user needs. So we let many signals share it. There are two different situations, and the question is about the difference.

## 2. What is multiplexing?

In **multiplexing**, the signals start at one place. A device called a **multiplexer** (MUX) combines them, sends them over one link, and a **demultiplexer** at the other end splits them again. Think of one truck carrying the parcels of many people between two depots. The sharing is arranged in advance at one point.

## 3. What is multiple access?

In **multiple access**, the users are in different places and each sends for itself, as mobile phones do around one tower. No single box combines them, so they mix in the channel itself. A set of rules must therefore tell each user which frequency, time slot or code to use, and users can join and leave. Think of many people sharing one road, each driving their own car.


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## 2. Answer

| | Multiplexing | Multiple access |
|---|---|---|
| What is shared | one link between two points | one medium (e.g. the air around a tower) by many separate users |
| Where the signals are combined | at one place, by a multiplexer (MUX) | in the channel itself; each user transmits from its own place |
| Control | one MUX/DEMUX pair, fixed allocation | a protocol assigns resources to users; users join and leave |
| Techniques | FDM, TDM, WDM | FDMA, TDMA, CDMA |
| Example | T-1/E-1 telephone trunks, FDM radio | mobile phone networks (GSM, CDMA), satellite access |

**Table 7:** Multiplexing versus multiple access

In short: multiplexing combines several signals at one point onto one link; multiple access lets several distant users share one medium. Each multiple-access method is a multiplexing idea applied to separate users.

## 3. Final Concise Answer

**(a)** Multilevel: combine slow inputs first, then multiplex with faster ones ($20 + 20 + 40 \to 80$ kbps). Multiple-slot: give a faster input several slots per frame in proportion to its rate (5, 10, 15 kbps $\to$ 1, 2, 3 slots $\to$ 30 kbps).

**(b)** $f_s = 400$ Hz; $m_p/L \leq 0.0025 m_p \Rightarrow L \geq 400 \Rightarrow n = 9$; $R_b = 10 \times 400 \times 9 = 36$ kbit/s; $B_T = 18$ kHz.

**(c)** FDMA: separate frequency bands, guard bands, hard capacity limit. CDMA: same wide band and time, separated by codes, soft capacity, needs power control, resists jamming.

**(d)** Requirements $\to$ channel study $\to$ Shannon limit $\to$ source coding $\to$ channel coding $\to$ modulation $\to$ multiplexing/multiple access $\to$ link budget $\to$ hardware $\to$ simulate and test; balance bandwidth, power and cost.

**(e)** Multiplexing shares one link at one point (MUX: FDM, TDM, WDM); multiple access shares a medium among separate users (FDMA, TDMA, CDMA).


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FORMULA SHEET

# EEE329

Basic Communication Engineering

# Formula Sheet

**Session 2021–22 paper** *Every formula behind the Third Year Second Semester Final Examination 2024*

## TOPICS


1 Information and channel capacity — 9 formulas

2 Channels and propagation — 4 formulas

3 Amplitude modulation — 11 formulas

4 Frequency modulation — 6 formulas

5 Sampling and PCM — 13 formulas

6 Digital modulation — 7 formulas

7 Multiplexing — 3 formulas


Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

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BACK OF THE COVER

# Before you use the sheet

## How to read a row

| | | | |
|---|---|---|---|
| **Information in a symbol** | $I(m_i) = \log_2 \frac{1}{P_i}$ bits | $P_i$ = probability of symbol $m_i$ | 1(a) |
| the name you search for | the formula, exactly as derived | every new symbol with its unit | link: opens the derivation |

## In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

# Contents

1 Information and channel capacity 4

2 Channels and propagation 5

3 Amplitude modulation 5

4 Frequency modulation 6

5 Sampling and PCM 6

6 Digital modulation 7

7 Multiplexing 8

8 Key formulas 9

9 Symbol index 10



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Conventions (Lathi and Ding, 4th ed.): $\omega = 2\pi f$; $m(t)$ is the message, $m_p$ its peak value and $B$ its bandwidth (Hz); $\mu$ is the AM modulation index and $\beta$ the FM one; $\mathrm{sinc}(x) = \sin x/x$; $\log_2$ gives bits; $S/N$ inside a formula is the plain power ratio, not dB. References point to the solutions of the 2021–22 paper.

**Columns:** Title · Equation · Notation · Ref. (reference).

## 1. Information and channel capacity:

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| Information in a symbol | $I(m_i) = \log_2 \dfrac{1}{P_i}$ bits | $P_i$ = probability of symbol $m_i$ | 1(a) |
| ★ **Entropy** | $H(m) = \sum_{i=1}^{n} P_i \log_2 \dfrac{1}{P_i}$ bits/symbol | $H$ = average information per symbol<br>$n$ = number of symbols | 1(a) |
| Bounds of entropy | $0 \le H(m) \le \log_2 n$ | maximum when all $P_i = 1/n$ | 1(a) |
| Binary entropy | $H(p) = -p \log_2 p - (1-p)\log_2(1-p)$ | $p$ = probability of one of two symbols; 1 bit at $p = 0.5$ | 1(a) |
| ★ **Information rate** | $R = r\,H$ bits/s | $r$ = symbol rate (symbols/s); needs $R \le C$ | 4(c) |
| Attenuation | $10 \log_{10} \dfrac{P_t}{P_r} = 20 \log_{10} \dfrac{V_t}{V_r}$ dB | $P_t$, $P_r$ = sent and received power (W)<br>$V_t$, $V_r$ = sent and received amplitude (V) | 1(c) |
| Signal-to-noise ratio | $\dfrac{S}{N} = \left(\dfrac{V_s}{V_n}\right)^2$ | $V_s$, $V_n$ = signal and noise amplitude (V), same resistance | 1(c) |
| ★ **Shannon capacity** | $C = B \log_2 \left(1 + \dfrac{S}{N}\right)$ bit/s | $C$ = largest error-free rate (bit/s)<br>$B$ = channel bandwidth (Hz) | 1(c) |
| SNR for a rate | $\dfrac{S}{N} = 2^{R/B} - 1$ | $R$ = wanted data rate (bit/s) | 1(c) |

**Table 1:** Information and channel capacity

## 2. Channels and propagation:

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| Distortionless output | $y(t) = k\,x(t - t_d)$ | $k$ = constant gain<br>$t_d$ = constant delay (s) | 1(d) |
| ★ **Distortionless channel** | $H(f) = k\,e^{-j2\pi f t_d}$ | $\lvert H(f)\rvert = k$ and<br>$\theta_h(f) = -2\pi f t_d$<br>over the signal band | 1(d) |
| Group delay | $t_g = -\frac{1}{2\pi}\frac{d\theta_h}{df} = t_d$ | $\theta_h$ = phase response (rad) | 1(d) |
| Radio horizon (LOS) | $d \approx 4.12\left(\sqrt{h_t} + \sqrt{h_r}\right)$ km | $h_t$, $h_r$ = antenna heights (m),<br>$\frac{4}{3}$-Earth radius | 1(b) |

**Table 2:** Channels and propagation

## 3. Amplitude modulation:

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| AM wave | $\varphi_{AM}(t) = [A + m(t)]\cos\omega_c t$ | $A$ = carrier amplitude<br>$\omega_c$ = carrier frequency (rad/s) | 3(d) |
| ★ **Modulation index** | $\mu = \frac{m_p}{A}$ | envelope detection needs<br>$A + m(t) \geq 0$, i.e. $\mu \leq 1$ | 3(d) |
| ★ **AM power** | $P_t = P_c\left(1 + \frac{\mu^2}{2}\right)$ | $P_t$ = total power, $P_c$ = carrier power (W) | 2(a) |
| ★ **AM antenna current** | $I_t = I_c\sqrt{1 + \frac{\mu^2}{2}}$ | $I_t$ = rms current with modulation (A)<br>$I_c$ = rms current of the carrier alone (A) | 2(a) |
| Envelope detector RC | $\frac{1}{\omega_c} \ll RC < \frac{1}{2\pi B}$ | $RC$ = load time constant (s) | 3(b) |
| ★ **Coherent detection with errors** | $e_o(t) = m(t)\cos(\Delta\omega\, t + \theta)$ | $\Delta\omega$ = local-carrier frequency error (rad/s)<br>$\theta$ = phase error (rad) | 2(b) |
| Costas I and Q outputs | $\frac{1}{2}m(t)\cos\theta,\ \frac{1}{2}m(t)\sin\theta$ | I and Q branch outputs after the LPF | 2(c) |
| Costas control signal | $\frac{1}{8}m^2(t)\sin 2\theta$ | after the narrow LPF $\approx \frac{1}{4}\overline{m^2}\,\theta$ for small $\theta$ | 2(c) |
| VSB signal | $\Phi_{VSB}(f) = [M(f + f_c) + M(f - f_c)]\,H_i(f)$ | $M(f)$ = message spectrum<br>$H_i(f)$ = VSB shaping filter | 2(d) |
| ★ **VSB filter condition** | $H_i(f + f_c) + H_i(f - f_c) = \text{constant},\ \lvert f\rvert \leq B$ | odd symmetry about $f_c$, half gain at $f_c$ | 2(d) |

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| VSB receiver filter | $H_o(f) = \dfrac{1}{H_i(f+f_c) + H_i(f-f_c)}$ | $H_o(f)$ = receiver LPF, for $\lvert f\rvert \le B$ | 2(d) |

**Table 3:** Amplitude modulation

## 4. Frequency modulation:

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| FM wave | $\varphi_{FM}(t) = A\cos[\omega_c t + k_f a(t)],\ a(t) = \int m(\tau)\,d\tau$ | $k_f$ = frequency sensitivity (rad/s per volt) | 3(a) |
| ★ **NBFM** | $\varphi_{NBFM}(t) \approx A\cos\omega_c t - A\,k_f\,a(t)\sin\omega_c t$ | valid for $\lvert k_f a(t)\rvert \ll 1$; bandwidth $\approx 2B$ | 3(a) |
| Frequency limits | $f_{max,min} = f_c \pm \Delta f$ | $\Delta f$ = peak frequency deviation (Hz) | 3(c) |
| Carrier swing | swing $= 2\Delta f$ | total frequency excursion (Hz) | 3(c) |
| ★ **FM modulation index** | $\beta = \dfrac{\Delta f}{f_m}$ | $f_m$ = modulating frequency (Hz) | 3(c) |
| Carson's rule | $B_T \approx 2(\Delta f + f_m)$ | $B_T$ = FM bandwidth (Hz) | 3(c) |

**Table 4:** Frequency modulation

## 5. Sampling and PCM:

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| Sampled spectrum | $\bar{G}(f) = \dfrac{1}{T_s}\sum_{n=-\infty}^{\infty} G(f - nf_s)$ | $G(f)$ = message spectrum<br>$T_s = 1/f_s$ = sampling interval (s) | 4(b) |
| ★ **Nyquist criterion** | $f_s \ge 2B$ | $f_s$ = sampling rate (samples/s) | 4(b) |
| Reconstruction LPF | $H(f) = T_s\,\mathrm{rect}\left(\dfrac{f}{2f_0}\right),\ B \le f_0 \le f_s - B$ | $f_0$ = filter cutoff (Hz) | 4(b) |
| ★ **Interpolation formula** | $g(t) = \sum_k g(kT_s)\,\mathrm{sinc}(2\pi Bt - k\pi)$ | for Nyquist sampling, $f_s = 2B$ | 4(b) |
| Step size | $\Delta = \dfrac{2m_p}{L}$ | $L$ = number of levels<br>largest error $\Delta/2 = m_p/L$ | 4(e) |
| Bits per sample | $L = 2^n,\ n = \log_2 L$ | $n$ = bits per code word | 4(e) |
| Quantization noise | $N_q = \dfrac{\Delta^2}{12} = \dfrac{m_p^2}{3L^2}$ | average noise power | 4(a) |


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| Title | Equation | Notation | Ref. |
|---|---|---|---|
| ★ **SQNR** | $\text{SQNR} = 3L^2 \dfrac{\overline{m^2}}{m_p^2}$ | $\overline{m^2}$ = message power; about $6.02n$ dB + const. | 4(a) |
| Peak signal to peak noise | $\dfrac{S_{peak}}{N_{peak}} = L^2$ | $3L^2$ with the average noise $\Delta^2/12$ | 4(e) |
| ★ **PCM bit rate** | $R_b = nf_s$ | $R_b$ = bit rate (bit/s) | 4(e) |
| ★ **Minimum bandwidth** | $B_T = \dfrac{R_b}{2} = nB$ | $nB$ holds at Nyquist sampling | 4(e) |
| μ-law compressor | $y = \dfrac{\ln(1+\mu\lvert m\rvert/m_p)}{\ln(1+\mu)}\,\text{sgn}(m)$ | $\mu$ = compression constant (255), not the AM index | 4(c) |
| Companded SQNR | $\text{SQNR} \approx \dfrac{3L^2}{[\ln(1+\mu)]^2}$ | almost independent of the signal power | 4(c) |

**Table 5:** Sampling and PCM

## 6. Digital modulation:

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| ASK and PSK bandwidth | $B = (1+d)S$ | $S$ = signal (baud) rate<br>$d$ = filter factor,<br>$0 \le d \le 1$ | 4(d) |
| FSK bandwidth | $B = (1+d)S + 2\Delta f$ | $\Delta f$ = half the spacing of $f_1$ and $f_2$ (Hz) | 4(d) |
| ★ **QAM signal** | $\varphi_{QAM}(t) = m_1(t)\cos\omega_c t + m_2(t)\sin\omega_c t$ | $m_1$, $m_2$ = the two messages | 5(b) |
| QAM with phase error | $m_1(t)\cos\theta - m_2(t)\sin\theta$ | in-phase output; $m_2$ leaks in (crosstalk) | 5(b) |
| ★ **DPSK encoder** | $q_k = I_k \oplus q_{k-1}$ | $I_k$ = data bit<br>$q_k$ = encoded bit | 5(c) |
| DPSK receiver | $y_k \propto a_k a_{k-1}$ | $a_k$ = polar symbol ($\pm 1$)<br>$y_k$ = LPF output; negative means 1 | 5(c) |
| Coherent BASK output | $dA\cos^2\omega_c t \to \dfrac{dA}{2}$ | $d$ = data (1 or 0); threshold $A/4$ | 5(d) |

**Table 6:** Digital modulation


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## 7. Multiplexing:

| Title | Equation | Notation | Ref. |
|---|---|---|---|
| ★ **TDM bit rate** | $R_b = N f_s n$ | $N$ = number of channels | 6(b) |
| Levels from an error limit | $\frac{m_p}{L} \leq \epsilon\, m_p \;\Rightarrow\; L \geq \frac{1}{\epsilon}$ | $\epsilon$ = allowed error as a fraction of $m_p$ | 6(b) |
| Synchronous TDM link | $R_{link} = \sum R_i$ | $R_i$ = input rates; multilevel or multiple-slot | 6(a) |

**Table 7:** Multiplexing


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## 8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

| Title | Equation |
|---|---|
| Entropy | $H(m) = \sum_{i=1}^{n} P_i \log_2 \dfrac{1}{P_i}$ bits/symbol |
| Information rate | $R = r\,H$ bits/s |
| Shannon capacity | $C = B \log_2 \left(1 + \dfrac{S}{N}\right)$ bit/s |
| Distortionless channel | $H(f) = k\,e^{-j2\pi f t_d}$ |
| Modulation index | $\mu = \dfrac{m_p}{A}$ |
| AM power | $P_t = P_c \left(1 + \dfrac{\mu^2}{2}\right)$ |
| AM antenna current | $I_t = I_c \sqrt{1 + \dfrac{\mu^2}{2}}$ |
| Coherent detection with errors | $e_o(t) = m(t)\cos(\Delta\omega\, t + \theta)$ |
| VSB filter condition | $H_i(f + f_c) + H_i(f - f_c) = \text{constant},\ \lvert f\rvert \le B$ |
| NBFM | $\varphi_{NBFM}(t) \approx A\cos\omega_c t - A\,k_f\,a(t)\sin\omega_c t$ |
| FM modulation index | $\beta = \dfrac{\Delta f}{f_m}$ |
| Nyquist criterion | $f_s \ge 2B$ |
| Interpolation formula | $g(t) = \sum_k g(kT_s)\,\mathrm{sinc}(2\pi B t - k\pi)$ |
| SQNR | $\text{SQNR} = 3L^2\,\dfrac{\overline{m^2}}{m_p^2}$ |
| PCM bit rate | $R_b = n f_s$ |
| Minimum bandwidth | $B_T = \dfrac{R_b}{2} = nB$ |
| QAM signal | $\varphi_{QAM}(t) = m_1(t)\cos\omega_c t + m_2(t)\sin\omega_c t$ |
| DPSK encoder | $q_k = I_k \oplus q_{k-1}$ |
| TDM bit rate | $R_b = N f_s n$ |



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## 9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

| Symbol | Meaning | First used in |
|---|---|---|
| $P_i$ | probability of symbol $m_i$ | Information in a symbol |
| $H$ | average information per symbol | Entropy |
| $n$ | number of symbols | Entropy |
| $p$ | probability of one of two symbols; 1 bit at $p = 0.5$ | Binary entropy |
| $r$ | symbol rate (symbols/s); needs $R \leq C$ | Information rate |
| $P_t$, $P_r$ | sent and received power (W) | Attenuation |
| $V_t$, $V_r$ | sent and received amplitude (V) | Attenuation |
| $V_s$, $V_n$ | signal and noise amplitude (V), same resistance | Signal-to-noise ratio |
| $C$ | largest error-free rate (bit/s) | Shannon capacity |
| $B$ | channel bandwidth (Hz) | Shannon capacity |
| $R$ | wanted data rate (bit/s) | SNR for a rate |
| $k$ | constant gain | Distortionless output |
| $t_d$ | constant delay (s) | Distortionless output |
| $\theta_h$ | phase response (rad) | Group delay |
| $h_t$, $h_r$ | antenna heights (m), $\frac{4}{3}$-Earth radius | Radio horizon (LOS) |
| $A$ | carrier amplitude | AM wave |
| $\omega_c$ | carrier frequency (rad/s) | AM wave |
| $P_t$ | total power, $P_c$ = carrier power (W) | AM power |
| $I_t$ | rms current with modulation (A) | AM antenna current |
| $I_c$ | rms current of the carrier alone (A) | AM antenna current |
| $RC$ | load time constant (s) | Envelope detector RC |
| $\Delta\omega$ | local-carrier frequency error (rad/s) | Coherent detection with errors |
| $\theta$ | phase error (rad) | Coherent detection with errors |
| $M(f)$ | message spectrum | VSB signal |
| $H_i(f)$ | VSB shaping filter | VSB signal |
| $H_o(f)$ | receiver LPF, for $\lvert f\rvert \leq B$ | VSB receiver filter |
| $k_f$ | frequency sensitivity (rad/s per volt) | FM wave |
| $\Delta f$ | peak frequency deviation (Hz) | Frequency limits |
| $f_m$ | modulating frequency (Hz) | FM modulation index |
| $B_T$ | FM bandwidth (Hz) | Carson's rule |
| $G(f)$ | message spectrum | Sampled spectrum |
| $T_s = 1/f_s$ | sampling interval (s) | Sampled spectrum |
| $f_s$ | sampling rate (samples/s) | Nyquist criterion |
| $f_0$ | filter cutoff (Hz) | Reconstruction LPF |
| $L$ | number of levels | Step size |

| Symbol | Meaning | First used in |
|---|---|---|
| $\overline{m^2}$ | message power; about $6.02n$ dB + const. | SQNR |
| $R_b$ | bit rate (bit/s) | PCM bit rate |
| $\mu$ | compression constant (255), not the AM index | μ-law compressor |
| $S$ | signal (baud) rate | ASK and PSK bandwidth |
| $d$ | filter factor, $0 \leq d \leq 1$ | ASK and PSK bandwidth |
| $m_1$, $m_2$ | the two messages | QAM signal |
| $I_k$ | data bit | DPSK encoder |
| $q_k$ | encoded bit | DPSK encoder |
| $a_k$ | polar symbol ($\pm 1$) | DPSK receiver |
| $y_k$ | LPF output; negative means 1 | DPSK receiver |
| $N$ | number of channels | TDM bit rate |
| $\epsilon$ | allowed error as a fraction of $m_p$ | Levels from an error limit |
| $R_i$ | input rates; multilevel or multiple-slot | Synchronous TDM link |


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